



ANKARA YILDIRIM BEYAZIT UNIVERSITY

GRADUATE SCHOOL OF NATURAL AND APPLIED SCIENCES

**OPTIMIZATION OF LOCATION ASSIGNMENT AND ON-CALL
PHARMACY PLANNING FOR NEWLY OPENING PHARMACIES**

Thesis by

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**OPTIMIZATION OF LOCATION ASSIGNMENT AND ON-CALL
PHARMACY PLANNING FOR NEWLY OPENING PHARMACIES**

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ETHICAL DECLARATION

I hereby declare that this project report is entirely my own work. I confirm that throughout all stages of this study, from planning to writing, I have complied with academic and ethical principles and have not engaged in any behavior that would violate copyright, patent rights, or academic integrity.

I further declare that all information, data, findings, and interpretations presented in this report have been obtained in accordance with scientific and ethical standards, and that all sources used in the preparation of this study have been appropriately acknowledged and cited.

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ÖZET

Yeni Açılan Eczaneler İçin Lokasyon Ataması ve Nöbetçi Eczane Planlamasının Optimizasyonu

Eczaneler, sağlık hizmetlerine erişimin sağlanmasında ve toplum sağlığının korunmasında önemli bir role sahiptir. Ancak eczanelerin mekânsal dağılımındaki dengesizlikler ve nöbet hizmetlerinin etkin planlanamaması, sağlık hizmetlerine erişimi olumsuz etkileyebilmektedir. Bu nedenle eczane yer seçimi ve nöbetçi eczane planlaması önemli karar problemleri arasında yer almaktadır.

Bu çalışmada, yeni açılacak eczanelerin yer seçimi ve talep noktalarına atanması problemi ile nöbetçi eczane planlama problemi bütünleşik bir yaklaşımla ele alınmıştır. Çalışmanın amacı, sağlık hizmetlerine erişilebilirliği artırmak, mekânsal eşitsizlikleri azaltmak ve nöbet hizmetlerinin sürdürülebilirliğini sağlamaktır. Uygulama alanı olarak Ağrı il merkezi seçilmiş ve analizlerde gerçek mekânsal veriler kullanılmıştır.

Çalışmanın ilk aşamasında, eczane yer seçimi ve talep atama problemi için çok amaçlı bir Karma Tamsayılı Doğrusal Programlama (MILP) modeli geliştirilmiştir. Modelde P-Median ve P-Center yaklaşımları birlikte kullanılarak erişilebilirlik, mekânsal adalet, sağlık kuruluşlarına yakınlık, kapasite kısıtları ve hizmet kapsama kriterleri değerlendirilmiştir. Sonuçlar Coğrafi Bilgi Sistemleri (CBS) yardımıyla analiz edilmiştir.

İkinci aşamada ise nöbet yükünün dengeli dağıtılması, ardışık nöbetlerin önlenmesi, dinlenme sürelerinin korunması ve her gün yeterli sayıda nöbetçi eczanenin hizmet vermesini sağlayan bir nöbetçi eczane çizelgeleme modeli geliştirilmiştir.

Elde edilen sonuçlar, önerilen yaklaşımın hem eczane yer seçimi kararlarında erişilebilirliği ve mekânsal eşitliği iyileştirdiğini hem de nöbet hizmetlerinin daha adil ve sürdürülebilir şekilde planlanmasına katkı sağladığını göstermektedir. Ayrıca geliştirilen modellerin farklı şehir ve bölgelerde uygulanabilecek karar destek araçları olduğu görülmüştür.

Anahtar Kelimeler: Eczane Yer Seçimi, Nöbetçi Eczane Çizelgeleme, Çok Amaçlı Optimizasyon, Karma Tamsayılı Doğrusal Programlama, P-Median, P-Center, Coğrafi Bilgi Sistemleri.

SUMMARY

Optimization of Location Assignment and On-Call Pharmacy Planning for Newly Opening Pharmacies

Pharmacies play a vital role in ensuring access to healthcare and protecting public health. However, imbalances in the spatial distribution of pharmacies and the inability to effectively plan on-call services can negatively impact access to healthcare. Therefore, pharmacy location selection and on-call pharmacy planning are among the important decision-making problems.

This study addresses the problem of selecting locations for newly opened pharmacies and assigning them to demand points, along with the problem of on-call pharmacy planning, using an integrated approach. The aim of the study is to increase accessibility to healthcare services, reduce spatial inequalities, and ensure the sustainability of on-call services. Ağrı city center was selected as the application area, and real and generated spatial inputs were used in the analyses.

In the first stage of the study, a multi-objective Mixed Integer Linear Programming (MILP) model was developed for the pharmacy location selection and demand assignment problem. Using P-Median and P-Center approaches together, the model evaluated accessibility, spatial justice, proximity to healthcare facilities, capacity constraints, and service coverage criteria. The results were analyzed using Geographic Information Systems (GIS).

In the second stage, an on-duty pharmacy scheduling model was developed that ensures balanced distribution of the duty load, prevention of consecutive seizures, protection of rest periods and the provision of a sufficient number of pharmacies on duty every day.

The results obtained show that the proposed approach both improves accessibility and spatial equality in pharmacy location decisions and contributes to more equitable and sustainable planning of on-call services. In addition, it has been seen that the developed models are decision support tools that can be applied in different cities and regions.

Keywords: Pharmacy Location Selection, On-Duty Pharmacy Scheduling, Multi-Objective Optimization, Mixed Integer Linear Programming, P-Median, P-Center, Geographic Information Systems.

SYMBOLS AND ABBREVIATIONS

Abbreviations

Abbreviation	Description
GIS	Geographic Information Systems
MILP	Mixed Integer Linear Programming
NWSM	Normalized Weighted Sum Method
MCDM	Multi-Criteria Decision Making
P-Median	P-Median Facility Location Model
P-Center	P-Center Facility Location Model
OR	Operations Research
AHP	Analytic Hierarchy Process

Sets and Indices

Symbol	Description
(i)	Demand point index
(j)	Candidate pharmacy location index
(k)	Pharmacy index
(t)	Day index
(I)	Set of demand points
(J)	Set of candidate pharmacy locations
(K)	Set of pharmacies
(T)	Set of planning days

Parameters

Symbol	Description
(w_i)	Demand weight of demand point i

Symbol	Description
(RW_i)	Vulnerability-adjusted demand weight of demand point i
(d_{ij})	Distance between demand point i and candidate location j
(c_j)	Opening cost of pharmacy at location j
(h_j)	Healthcare infrastructure proximity score of location j
(P)	Number of pharmacies to be opened
(D_{max})	Maximum service distance threshold
(M)	Large positive constant (Big-M parameter)
(N)	Total number of pharmacies
(R)	Required number of duty pharmacies per day
(α)	Weight of accessibility objective
(β)	Weight of equity objective
(γ)	Weight of opening cost objective
(δ)	Weight of healthcare proximity objective

Decision Variables

Symbol	Description
(x_j)	1 if a pharmacy is opened at location j, 0 otherwise
(y_{ij})	1 if demand point i is assigned to pharmacy j, 0 otherwise
(u_i)	1 if demand point i is uncovered, 0 otherwise
(Z)	Overall objective function value
(Z_{PM})	P-Median objective value
(Z_{PC})	P-Center objective value
(Z_{Cost})	Opening cost objective value
(Z_{Health})	Healthcare proximity objective value

Symbol	Description
--------	-------------

(W)	Maximum weighted assignment distance
(z_{kt})	1 if pharmacy (k) is assigned duty on day (t), 0 otherwise
(L_k)	Total duty workload assigned to pharmacy (k)
(L_{max})	Maximum duty workload among pharmacies
(L_{min})	Minimum duty workload among pharmacies
(Δ)	Workload imbalance measure

TABLES

Table 1. Sparse MILP Computational Performance Metrics

Table 2. Accessibility and Spatial Equity Performance Indicators

FIGURES

Figure 1. Proposed Two-Stage Pharmacy Location and Duty Scheduling Framework

Figure 2. Spatial Assignment of Demand Points to Selected Pharmacies

Figure 3. Optimized Pharmacy Location and Accessibility Map for Ağrı City Center

1. INTRODUCTION

Access to healthcare services is one of the most fundamental components of social welfare, public health protection, and sustainable urban development. Among healthcare facilities, community pharmacies occupy a unique position because they constitute the most accessible and frequently utilized healthcare service points within everyday life. Beyond their traditional role in medication dispensing, pharmacies provide patient counseling, preventive healthcare services, chronic disease support, medication management, and pharmaceutical guidance. Consequently, the effectiveness of pharmacy networks directly influences healthcare accessibility, treatment continuity, and population health outcomes.

In recent decades, healthcare accessibility has increasingly been recognized as a multidimensional concept extending beyond the simple presence of healthcare facilities. Contemporary healthcare planning emphasizes not only efficiency but also equity, spatial justice, service continuity, and accessibility for vulnerable populations. Under this perspective, healthcare resources should be distributed in a manner that ensures fair access for all citizens regardless of their socioeconomic status, geographic location, age structure, or mobility limitations. Therefore, healthcare facility planning has evolved from a purely operational problem into a broader socio-spatial planning challenge.

Community pharmacies are particularly important in this context because they represent the first point of contact between citizens and the healthcare system. Elderly individuals, patients with chronic diseases, low-income households, disabled populations, and residents of peripheral neighborhoods often depend heavily on nearby pharmaceutical services. Long travel distances, transportation difficulties, and inadequate service coverage may lead to medication non-adherence, treatment interruptions, delayed healthcare utilization, and reduced quality of life. Consequently, the spatial distribution of pharmacies directly affects healthcare equity and accessibility.

In Türkiye, the establishment and relocation of pharmacies are regulated under population-based criteria introduced through the Regulation on Pharmacists and Pharmacies. While these regulations were designed to prevent excessive clustering and ensure a balanced distribution of pharmaceutical services, practical implementation has revealed several limitations. Population thresholds alone are often insufficient to

represent the heterogeneous accessibility conditions experienced across different urban environments. Districts with similar population sizes may exhibit substantially different socioeconomic characteristics, transportation opportunities, age distributions, healthcare needs, and accessibility barriers.

As a result, compliance with population-based regulations does not necessarily guarantee equitable access to pharmaceutical services. Pharmacies tend to concentrate in commercially attractive and densely populated urban centers where economic opportunities are stronger. Conversely, peripheral districts, newly developing neighborhoods, rural settlements, and socioeconomically disadvantaged communities frequently experience lower levels of pharmaceutical accessibility despite often exhibiting greater healthcare vulnerability. These patterns create spatial inequalities that may negatively affect healthcare outcomes and social welfare.

The growing concern regarding unequal healthcare accessibility has led researchers to focus on concepts such as spatial justice, healthcare equity, and pharmacy deserts. Pharmacy deserts refer to areas where residents experience inadequate access to pharmaceutical services due to excessive travel distances, weak transportation infrastructure, or insufficient service density. These regions often overlap with populations characterized by higher levels of socioeconomic vulnerability, aging demographics, limited mobility, and increased healthcare needs. Consequently, reducing pharmacy desert formation has become an important objective in modern healthcare planning.

Despite increasing attention to healthcare accessibility problems, much of the existing literature remains primarily descriptive. Geographic Information Systems (GIS)-based studies have successfully identified accessibility inequalities, spatial clustering patterns, and underserved regions. However, most studies stop at diagnosing the problem rather than proposing prescriptive solutions capable of guiding future pharmacy placement decisions. Similarly, many optimization-based studies focus exclusively on efficiency-oriented objectives such as minimizing average travel distance, often neglecting equity considerations and vulnerability-sensitive planning principles.

Another important limitation concerns the separation between strategic and operational healthcare planning. Traditional facility location studies typically determine where facilities should be established but rarely consider how these facilities

should operate during emergency periods, weekends, public holidays, or nighttime service requirements. Consequently, strategic accessibility and operational service continuity are often evaluated independently despite being strongly interconnected components of healthcare accessibility.

To address these gaps, this study develops an integrated, equity-oriented, and optimization-based decision-support framework for pharmacy planning. The proposed framework combines strategic facility location decisions with operational duty-pharmacy scheduling within a unified analytical structure. The study is implemented using real, generated spatial and demographic data from Ağrı city center, providing an empirical environment for evaluating accessibility-oriented healthcare planning approaches under realistic urban conditions.

Ağrı represents a particularly suitable case study due to its heterogeneous urban structure, peripheral residential areas, varying accessibility conditions, and socioeconomic diversity. The coexistence of densely populated central districts and relatively underserved peripheral neighborhoods creates a meaningful environment for analyzing the interaction between efficiency, equity, and healthcare accessibility. Furthermore, the presence of healthcare infrastructure nodes such as Ağrı Training and Research Hospital enables the examination of healthcare integration within pharmacy planning decisions.

The proposed framework is formulated as a multi-objective Mixed Integer Linear Programming (MILP) model. Unlike traditional facility location models that rely solely on distance minimization, the proposed approach simultaneously considers multiple planning objectives. Specifically, the model integrates the classical P-median and P-center formulations within a normalized objective structure. The P-median component aims to minimize weighted average accessibility distances, thereby improving overall system efficiency. The P-center component minimizes maximum accessibility distances, protecting geographically disadvantaged demand points and strengthening spatial equity.

In addition to accessibility objectives, the model incorporates deterministic opening cost proxies and healthcare facility proximity considerations. Through these components, newly selected pharmacy locations are encouraged to maintain stronger spatial integration with existing healthcare infrastructure while preserving operational feasibility. Furthermore, vulnerability-sensitive demand structures, weighted

accessibility measures, and realistic capacity constraints are incorporated into the model to better represent heterogeneous healthcare needs across the urban system.

To improve practical applicability, the framework also integrates Geographic Information Systems (GIS) into the analytical process. GIS-based visualization tools are employed to generate accessibility maps, assignment structures, service coverage analyses, pharmacy workload distributions, and spatial decision-support outputs. These visual representations improve the interpretability of optimization results and facilitate communication with healthcare planners and policymakers.

Beyond strategic location decisions, the study incorporates an operational duty-pharmacy scheduling layer. This second-stage framework evaluates how selected pharmacies can provide continuous pharmaceutical services during emergency and after-hours periods. Through the integration of strategic location-allocation decisions and operational scheduling mechanisms, the proposed framework offers a more comprehensive representation of real-world pharmaceutical service systems.

The findings of this research are expected to contribute both theoretically and practically to healthcare planning literature. From a theoretical perspective, the study extends classical facility location theory by integrating efficiency, equity, healthcare infrastructure proximity, and operational service continuity within a unified optimization framework. From a practical perspective, it provides a prescriptive decision-support tool capable of assisting policymakers, municipalities, healthcare authorities, and professional organizations in designing more accessible, equitable, and sustainable pharmaceutical service networks.

Ultimately, this study demonstrates how multi-objective optimization, spatial analysis, and healthcare planning principles can be combined to support evidence-based pharmacy location decisions. By moving beyond purely descriptive analyses and transforming healthcare accessibility into a formal optimization problem, the proposed framework offers a practical pathway toward more balanced and socially responsive pharmaceutical service distribution in Türkiye.

2.LITERATURE

2.1. Community Pharmacies, Access and Healthcare Accessibility

Access to healthcare has increasingly been conceptualized as a complex, multidimensional socio-spatial phenomenon that extends far beyond the mere physical presence or quantitative availability of medical resources within a geographical territory. Contemporary public health research, urban economics, and social geography emphasize that accessibility is a heterogeneous construct shaped not only by travel distance but also by socioeconomic conditions, institutional arrangements, transportation infrastructure, healthcare literacy, and individuals' physical and cognitive capacities to engage with healthcare services. In this context, healthcare accessibility is commonly described through multiple interconnected dimensions, including availability, affordability, acceptability, accommodation, and physical accessibility (Adepoju et al., 2023; Chen et al., 2024). Consequently, modern healthcare planning increasingly adopts a broader perspective that recognizes accessibility as a central determinant of public health outcomes rather than a purely logistical concern.

This transformation reflects a significant shift in healthcare planning philosophy. Traditional approaches primarily focused on the quantitative supply of healthcare resources, whereas contemporary frameworks increasingly emphasize healthcare equity, social inclusion, and spatial justice. Within this perspective, healthcare services should not only exist within a region but should also be distributed in a manner that enables all population groups to access them effectively regardless of their socioeconomic status, age, physical mobility, or geographic location (Guagliardo, 2004; Delamater, 2013). Accessibility therefore represents a critical interface between healthcare infrastructure and social welfare, influencing both individual well-being and broader population health outcomes.

The concept of spatial justice has become particularly influential in discussions of healthcare accessibility. Spatial justice perspectives argue that healthcare resources should be distributed fairly across geographic space to prevent systematic disadvantages among vulnerable populations. When healthcare infrastructure is concentrated within economically attractive urban centers while peripheral and disadvantaged communities remain underserved, spatial inequalities emerge that may

reinforce existing socioeconomic disparities. Consequently, healthcare accessibility is increasingly viewed not only as an operational objective but also as a normative public policy concern directly associated with social equity and inclusive urban development (Delamater, 2013; Murray, 2015).

Within this broader healthcare accessibility framework, community pharmacies occupy a distinctive and strategically important position. Compared with hospitals, specialized healthcare centers, and appointment-based clinical services, pharmacies generally provide highly accessible, low-threshold healthcare services. Citizens can obtain medications, refill prescriptions, receive pharmaceutical counseling, access preventive healthcare information, and seek guidance regarding medication use without lengthy administrative procedures or appointment requirements. This accessibility enables pharmacies to function as frontline healthcare access points within everyday life.

Empirical evidence indicates that interactions between citizens and community pharmacists often occur more frequently than interactions with primary care physicians, particularly in healthcare systems characterized by long waiting times, overloaded healthcare facilities, or limited physician accessibility (Dhital et al., 2022; Grootendorst, 2022). As a result, pharmacies serve as important intermediaries connecting formal healthcare institutions with community-level healthcare needs. Their role extends beyond medication dispensing and contributes directly to healthcare continuity, preventive care, chronic disease management, and patient education.

The functional scope of community pharmacies has expanded substantially during the past several decades. Modern pharmacies increasingly provide a variety of extended healthcare services, including vaccination programs, medication therapy management, chronic disease monitoring, smoking cessation programs, preventive health counseling, and public health education initiatives. These expanded responsibilities are particularly valuable for elderly populations, patients with chronic diseases, low-income households, and individuals requiring continuous access to healthcare services (Kevrekidis et al., 2018; Han et al., 2025). Consequently, the spatial accessibility of pharmacies directly affects medication adherence, treatment continuity, disease management outcomes, and overall population health.

The critical role of pharmacies became even more apparent during the COVID-19 pandemic. Throughout the pandemic period, pharmacies functioned as essential

healthcare access points, maintaining medication continuity, supporting public health communication, distributing preventive healthcare information, and alleviating pressure on hospitals and emergency healthcare facilities. This experience demonstrated the importance of maintaining resilient, geographically balanced, and operationally sustainable pharmacy networks capable of supporting healthcare systems during periods of extraordinary demand and uncertainty (Dhital et al., 2022; Grootendorst, 2022).

From a spatial accessibility perspective, uneven pharmacy distributions may generate significant healthcare inequalities. Market-driven location decisions often favor densely populated commercial districts characterized by strong purchasing power, high pedestrian traffic, and attractive economic conditions. In contrast, peripheral neighborhoods, rural settlements, newly developing urban areas, and socioeconomically disadvantaged communities frequently experience lower levels of pharmaceutical accessibility despite often exhibiting greater healthcare needs (Pednekar & Peterson, 2018; Mutlu et al., 2023). Such disparities may create substantial differences in healthcare utilization opportunities among population groups.

Numerous studies demonstrate that long travel distances, inadequate transportation infrastructure, mobility limitations, and fragmented healthcare service networks disproportionately affect elderly individuals, disabled populations, low-income households, and socially vulnerable communities. These barriers frequently contribute to delayed healthcare utilization, reduced medication adherence, interruptions in chronic disease management, and increased hospitalization risks (Pednekar & Peterson, 2018; Chen et al., 2024). Therefore, pharmacy accessibility should be considered a critical determinant of healthcare equity and social welfare rather than merely a transportation-related issue.

Healthcare continuity represents another important dimension of pharmacy accessibility. Access barriers may not only affect immediate healthcare utilization but may also influence long-term treatment adherence and preventive healthcare behaviors. Patients who require regular medication use often depend on repeated access to pharmaceutical services. When obtaining medications requires excessive travel effort or transportation costs, treatment continuity may gradually deteriorate, increasing healthcare vulnerability and reducing overall health outcomes.

Consequently, maintaining adequate pharmacy accessibility contributes directly to sustainable healthcare delivery and long-term population health improvement.

The economic sustainability of pharmacy operations further complicates healthcare planning decisions. Community pharmacies simultaneously function as healthcare providers and private commercial enterprises operating within regulated healthcare systems. This dual structure creates an inherent tension between profit-oriented location strategies and socially desirable accessibility outcomes. While concentration in commercially attractive districts may improve financial performance and operational efficiency, it may simultaneously reduce healthcare accessibility for populations residing in peripheral or disadvantaged areas (Grootendorst, 2022; Mutlu et al., 2023). Balancing these competing objectives therefore represents a major challenge for healthcare planners and policymakers.

Another important characteristic of healthcare accessibility is its dynamic nature. Accessibility conditions continuously evolve due to urban expansion, demographic change, population aging, migration patterns, transportation developments, economic restructuring, and changes in healthcare demand. Consequently, accessibility should not be viewed as a static phenomenon. Healthcare planning frameworks must be capable of adapting to changing spatial and demographic conditions while maintaining equitable access to healthcare services across diverse population groups (Chen et al., 2024; Han et al., 2025).

These complexities have encouraged researchers to conceptualize pharmacy planning as a formal decision problem rather than an outcome of unregulated market processes or static demographic regulations. Within the Operations Research literature, healthcare accessibility is increasingly approached through optimization-based methodologies that simultaneously evaluate efficiency, equity, operational feasibility, cost considerations, and healthcare infrastructure integration. Such approaches provide systematic mechanisms for identifying balanced solutions capable of supporting evidence-based healthcare planning and policy development (Murray, 2015; Özceylan et al., 2021).

Recent healthcare planning paradigms therefore move beyond simple average-distance measures and increasingly incorporate vulnerability-sensitive indicators, worst-case accessibility conditions, service continuity considerations, workload balancing structures, and equity-oriented performance metrics. This evolution reflects

a broader transition from purely efficiency-driven planning toward more inclusive and socially responsive healthcare accessibility frameworks (Han et al., 2025; Chen et al., 2024).

Recent advances in Geographic Information Systems (GIS), spatial analytics, and healthcare optimization have further strengthened the ability of researchers and policymakers to evaluate accessibility patterns. GIS-supported healthcare planning enables detailed visualization of service coverage areas, accessibility inequalities, demand-service mismatches, and spatial clustering patterns. When integrated with optimization models, GIS tools improve both the interpretability and practical applicability of healthcare planning decisions by providing intuitive representations of complex spatial relationships.

In addition to their direct healthcare functions, community pharmacies contribute significantly to the resilience and responsiveness of healthcare systems. During public health emergencies, natural disasters, supply chain disruptions, or seasonal disease outbreaks, pharmacies often serve as decentralized healthcare access points capable of maintaining service continuity while reducing pressure on hospitals and primary healthcare facilities. Their distributed structure allows healthcare systems to maintain flexibility and responsiveness during periods of elevated demand (Dhital et al., 2022; Grootendorst, 2022).

The strategic importance of pharmacy accessibility has therefore attracted growing attention from public health authorities, healthcare planners, and policymakers worldwide. As populations continue to age and healthcare expenditures increase, equitable access to pharmaceutical services has become both a healthcare objective and a socioeconomic necessity. Research suggests that improving access to primary healthcare services contributes not only to improved health outcomes but also to reductions in avoidable hospital admissions, emergency department congestion, and long-term healthcare expenditures (Adepoju et al., 2023; Delamater, 2013).

From a planning perspective, pharmacy accessibility represents a multidimensional challenge involving the interaction of demographic, socioeconomic, geographic, transportation-related, and healthcare infrastructure factors. Demand for pharmaceutical services is rarely distributed uniformly across urban environments. Instead, healthcare needs vary substantially according to population density, age composition, income levels, healthcare vulnerability, transportation accessibility, and

disease prevalence. Consequently, effective pharmacy planning requires analytical frameworks capable of simultaneously evaluating multiple dimensions of healthcare demand and accessibility (Chen et al., 2024; Han et al., 2025).

Recent developments in healthcare operations research increasingly advocate treating accessibility as a measurable and optimizable system characteristic. Rather than evaluating accessibility after facilities have been established, optimization-based planning approaches incorporate accessibility considerations directly into facility location decisions. Through mathematical optimization, healthcare planners can identify facility configurations that balance efficiency, equity, operational feasibility, and long-term sustainability while explicitly accounting for resource limitations and spatial constraints (Murray, 2015; Daskin, 2013; Özceylan et al., 2021).

Consequently, community pharmacy planning has evolved into a multidisciplinary research area positioned at the intersection of public health, healthcare management, urban planning, geography, operations research, and spatial analytics. Understanding pharmacy accessibility therefore requires not only examining healthcare service provision but also exploring the analytical frameworks capable of transforming accessibility objectives into practical and implementable planning decisions. This perspective provides the conceptual foundation for the facility location, accessibility modeling, and optimization approaches discussed in the following sections.

In summary, community pharmacies play a fundamental role in shaping healthcare accessibility, treatment continuity, public health outcomes, and spatial equity. Their distribution influences not only individual healthcare experiences but also the broader effectiveness, resilience, and fairness of healthcare systems. These characteristics make pharmacy planning a complex and multidimensional decision problem requiring integrated, accessibility-oriented, equity-sensitive, and operationally sustainable analytical frameworks supported by Geographic Information Systems, advanced spatial analysis, and multi-objective optimization methodologies (Adepoju et al., 2023; Grootendorst, 2022; Murray, 2015).

2.2. Pharmacy Deserts and Socioeconomic Vulnerability

A central concept in the international literature on healthcare accessibility optimization is that of the pharmacy desert, which refers to geographically distinct urban or regional patches characterized by limited, highly obstructed, or completely

inadequate access to essential pharmaceutical services due to physical distance, insufficient service density, weak transit infrastructure, or compounded socioeconomic barriers (Pednekar & Peterson, 2018; Catalano et al., 2024). Conceptually derived from the broader analytical framework of “food deserts,” the term emphasizes that regional healthcare accessibility inequalities are not accidental, temporary, or self-correcting phenomena; rather, they represent structural outcomes deeply embedded within urban, economic, infrastructural, and regional development systems. These spatial voids do not manifest in a vacuum; they are the physical manifestations of spatial sorting processes where institutional neglect and market logic overlap to push crucial therapeutic assets away from low-income or geographically isolated neighborhoods.

The concept of pharmacy deserts has become increasingly important within healthcare planning literature because it links spatial accessibility problems directly with social vulnerability and healthcare inequality. In this framework, accessibility is evaluated not only through physical distance but also through the ability of individuals and communities to effectively obtain pharmaceutical services under real-world conditions. Consequently, pharmacy deserts represent multidimensional spaces where healthcare disadvantage, social exclusion, and spatial inequality intersect simultaneously. The emergence of a pharmacy desert is an indicator of a breakdown in distributive justice, demonstrating that the baseline spatial configuration of a city fails to sustain a safe environment for its most exposed residents.

Pharmacy deserts are typically identified using spatial indicators such as distance to the nearest pharmacy, travel-time thresholds, pharmacy-to-population ratios, healthcare service density, and service coverage measurements. However, recent studies increasingly argue that geographic proximity alone does not adequately capture real-world healthcare accessibility conditions. Transportation availability, mobility limitations, digital exclusion, economic constraints, disability prevalence, healthcare literacy, and institutional barriers significantly shape effective healthcare access (Chen et al., 2024; Han et al., 2025). Accordingly, pharmacy deserts should not be conceptualized merely as physical voids within urban systems, but rather as multidimensional spaces of structural disadvantage and healthcare vulnerability. A neighborhood may reside within short geometric proximity to a medical cluster, yet

remain functionally isolated if severed by rapid transit corridors, industrial zones, topographically challenging terrains, or highly restrictive security perimeters.

Empirical evidence consistently demonstrates that elderly individuals, low-income households, rural populations, disabled individuals, and communities with weak transportation infrastructure are disproportionately affected by reduced pharmacy accessibility. Pednekar and Peterson (2018) demonstrate that longer travel distances are directly associated with lower medication adherence among older adults, leading to interruptions in chronic disease management, delayed treatments, and increased hospitalization risks. These findings reveal the cumulative nature of accessibility inequalities, whereby spatial disadvantage gradually transforms into long-term healthcare deterioration and reduced quality of life. The friction of distance acts as an economic and physical drain, converting routine therapeutic regimes into exhausting, high-cost exercises for citizens possessing compromised cognitive or ambulatory capacities.

Recent studies further emphasize that pharmacy accessibility problems are closely connected to broader patterns of social and economic marginalization. Catalano et al. (2024), for instance, identify strong associations between pharmacy deserts and indicators of social vulnerability such as unemployment, income deprivation, disability prevalence, and healthcare inequality. Their findings suggest that market-driven healthcare service distributions tend to reproduce existing socioeconomic inequalities rather than reduce them. Commercially attractive areas continue attracting healthcare investments, while already disadvantaged regions remain structurally underserved. This process accelerates a structural cycle where underfunded communities face a compounding loss of retail and medical infrastructure, which deepens their vulnerability to chronic and infectious diseases.

This structural imbalance becomes particularly significant in healthcare systems where pharmaceutical services are partially shaped by market dynamics. Pharmacies naturally tend to cluster in economically advantageous districts characterized by higher purchasing power, stronger commercial activity, and dense population structures. In contrast, peripheral urban areas, rural settlements, and socioeconomically vulnerable neighborhoods often remain underserved despite exhibiting higher healthcare needs. This mismatch between healthcare need and service availability exposes the absolute public health failure of purely market-

oriented allocation mechanisms, illustrating that capital investment operates on profit maximization vectors that rarely align with the biological demands of a regional population.

Importantly, vulnerability itself is not a static condition. Healthcare vulnerability continuously evolves in response to demographic aging, migration flows, urban expansion, transportation transformations, economic fluctuations, and public policy interventions. Consequently, static accessibility analyses may underestimate emerging risks and fail to identify future accessibility inequalities. To address this limitation, recent studies increasingly propose vulnerability indices, dynamic accessibility measures, predictive healthcare models, and GIS-supported spatial monitoring systems capable of identifying regions likely to become pharmacy deserts over time (Han et al., 2025; Chen et al., 2024).

Another important dimension discussed in the literature concerns the relationship between pharmacy deserts and healthcare resilience. Regions with limited pharmacy accessibility are often more vulnerable during healthcare crises, emergency situations, or disruptions within healthcare supply chains. The COVID-19 pandemic further highlighted that communities lacking proximate pharmaceutical access experienced greater difficulties in maintaining medication continuity, preventive healthcare communication, and basic healthcare support services. These observations reinforced the importance of maintaining resilient and spatially balanced pharmaceutical service networks within modern urban matrices to guarantee deep civilian defense structures.

The concept of pharmacy deserts also carries strong normative and ethical implications. It reframes limited pharmaceutical access not simply as a logistical inefficiency but as a form of social injustice and structural healthcare inequality. From this perspective, unequal access to essential pharmaceutical services constitutes a violation of fundamental healthcare rights, particularly for populations already affected by socioeconomic disadvantage and healthcare vulnerability (Pednekar & Peterson, 2018; Catalano et al., 2024). Therefore, healthcare accessibility problems should not be evaluated solely through efficiency metrics, but also through broader principles of distributive justice and equitable healthcare provision.

Urban and regional studies further indicate that pharmacy deserts frequently co-occur with other forms of spatial deprivation, including inadequate access to hospitals, primary healthcare facilities, supermarkets, public transportation systems, and social

services. This spatial co-location of disadvantages generates cumulative exclusionary dynamics that reinforce healthcare disparities and reduce social mobility opportunities across disadvantaged communities (Mutlu et al., 2023; Chen et al., 2024). In many cases, individuals living within pharmacy deserts simultaneously experience transportation poverty, digital exclusion, limited healthcare literacy, and economic vulnerability, intensifying overall healthcare accessibility barriers.

In the Turkish context, similar patterns of accessibility inequality have increasingly attracted attention within healthcare accessibility and spatial planning literature. Although population-based pharmacy regulations aim to establish balanced pharmaceutical service distribution, they often fail to fully capture heterogeneous vulnerability conditions across urban and regional environments. Consequently, compliance with demographic regulations does not necessarily guarantee equitable healthcare accessibility, particularly in rapidly growing metropolitan regions and peripheral districts.

Despite the growing recognition of pharmacy deserts and healthcare accessibility inequalities, most existing studies remain primarily descriptive and diagnostic. They successfully identify where accessibility problems exist but generally do not provide formal analytical mechanisms for eliminating or minimizing these inequalities systematically. This limitation highlights the necessity of prescriptive optimization frameworks capable of embedding vulnerability-sensitive criteria directly into healthcare planning and decision-making processes (Murray, 2015; Özceylan et al., 2021).

Recent developments in healthcare optimization literature increasingly advocate for integrating accessibility inequalities directly into mathematical decision structures rather than treating them solely as post hoc evaluation criteria. In this perspective, optimization models are not merely tools for minimizing travel distances or operational costs, but mechanisms for systematically improving healthcare equity and reducing structural accessibility disadvantages.

In the present study, the concept of pharmacy deserts serves as one of the foundational motivations underlying the proposed equity-oriented optimization framework, applied directly to the urban context of Ağrı city center. Rather than evaluating pharmacy deserts only after spatial allocation decisions are completed, vulnerability-sensitive accessibility conditions, bounded service thresholds (meters),

and strict infrastructure integration parameters are embedded directly into the mathematical structure of the proposed model.

Enforcing a strict geographic service perimeter of 3000.0 meters within the optimization core prevents the solver from generating distant assignments that look efficient on paper but cause physical exclusion for peripheral residents. Through this integrated approach, the study seeks not only to improve average accessibility performance but also to proactively mitigate the risk of pharmacy desert propagation within the disadvantaged peripheral neighborhoods of the city, forcing the mathematical programming matrix to divert optimization resources toward structurally isolated communities.

2.3. Socioeconomic Determinants of Pharmacy Distribution

The spatial distribution of pharmacies is not random; rather, it is shaped by a complex interaction of socioeconomic, demographic, infrastructural, and spatial factors. While geographic proximity is often treated as the primary indicator of healthcare accessibility, empirical research consistently demonstrates that socioeconomic conditions exert an equally, if not more, decisive influence on both the demand for pharmaceutical services and the feasibility of their provision (Catalano et al., 2024; Mutlu et al., 2023). Consequently, pharmacy accessibility should be interpreted not solely as a spatial phenomenon, but also as a socioeconomic and structural healthcare planning issue.

In healthcare accessibility literature, socioeconomic determinants are increasingly recognized as critical variables influencing both healthcare utilization and service distribution patterns. Accessibility inequalities emerge not only because of physical distance between individuals and healthcare facilities, but also because of differences in income, mobility, education, transportation access, demographic structure, healthcare literacy, and social vulnerability. These factors collectively shape the ability of populations to effectively access pharmaceutical services under real-world conditions. Therefore, a geographically nearby pharmacy does not necessarily imply equitable access if the surrounding population lacks the economic, social, or infrastructural resources required to utilize the service effectively.

Income level constitutes one of the most influential determinants of pharmacy location patterns. Higher-income neighborhoods tend to attract larger numbers of

pharmacies due to stronger purchasing power, greater commercial stability, higher rates of healthcare utilization, and more predictable consumption behavior. In contrast, low-income regions frequently experience reduced pharmacy density not because of lower healthcare need, but because of weaker commercial incentives and lower expected profitability (Grootendorst, 2022; Chen et al., 2024). This structural mismatch between healthcare need and market-driven service provision generates persistent spatial inequalities across urban and regional systems.

Furthermore, economically disadvantaged populations often face multiple overlapping healthcare barriers simultaneously. Limited financial resources may reduce transportation flexibility, medication affordability, healthcare continuity, and preventive healthcare utilization. Consequently, low-income communities frequently experience cumulative healthcare disadvantages even when healthcare facilities exist within moderate geographic proximity. Healthcare accessibility therefore depends not only on location but also on the socioeconomic capacity of individuals to convert spatial availability into actual healthcare utilization.

Age structure further amplifies accessibility inequalities. Elderly populations generally exhibit higher pharmaceutical demand due to chronic diseases, multimorbidity, long-term medication dependency, and increased healthcare utilization frequency. However, these populations often reside in peripheral urban districts, low-density settlements, or economically disadvantaged areas where healthcare service density remains limited. Efficiency-oriented location strategies frequently fail to ensure acceptable accessibility conditions for such groups because transportation burdens and maximum travel distances remain disproportionately high (Pednekar & Peterson, 2018; Han et al., 2025).

Population aging also increases the operational importance of proximate healthcare access. Elderly individuals often require repetitive and continuous interaction with pharmacies, making accessibility continuity particularly important for maintaining treatment adherence and healthcare stability. Inadequate pharmacy accessibility may therefore contribute not only to reduced healthcare utilization but also to broader declines in health outcomes and quality of life among aging populations.

Transportation infrastructure represents another critical determinant of effective healthcare accessibility. Accessibility is not solely determined by geographic distance but also depends on the availability, affordability, reliability, and safety of

transportation systems. Individuals without access to private vehicles or residing in regions characterized by weak public transportation networks frequently experience reduced healthcare accessibility even when pharmacies appear geographically close (Chen et al., 2024; Mutlu et al., 2023). Consequently, transportation inequality often translates directly into healthcare inequality.

This issue becomes particularly important in rapidly expanding metropolitan areas and developing medium-sized cities characterized by uneven urban development patterns and fragmented transportation networks. Peripheral settlements may experience substantial accessibility challenges due to weak transportation connectivity, inadequate pedestrian infrastructure, and excessive travel times despite moderate geographic distances. As a result, transportation disadvantage becomes an important mechanism through which healthcare accessibility inequalities emerge and persist.

Education and healthcare literacy further influence effective access to pharmaceutical services. Lower levels of education may restrict individuals' ability to navigate healthcare systems, interpret medical information, access digital healthcare platforms, understand medication instructions, or effectively utilize healthcare services (Catalano et al., 2024). Consequently, individuals may experience practical healthcare exclusion even when formal service availability appears adequate. Healthcare accessibility therefore contains important informational and cognitive dimensions in addition to purely spatial considerations.

The growing digitalization of healthcare systems has introduced new forms of accessibility inequality. Elderly populations, low-income households, and individuals with limited digital literacy may encounter difficulties using online appointment systems, electronic prescription platforms, telemedicine services, and digital healthcare applications. These barriers may intensify healthcare vulnerability among already disadvantaged populations and demonstrate that technological innovation alone cannot guarantee equitable healthcare access.

From an urban geography perspective, pharmacy location patterns are embedded within broader processes of spatial segregation, cumulative advantage, and uneven urban development. Commercial investments naturally tend to concentrate around affluent populations, dense commercial corridors, and economically dynamic urban districts, thereby reinforcing existing spatial privileges. Meanwhile,

socioeconomically disadvantaged neighborhoods frequently experience lower healthcare investment intensity and weaker service accessibility (Grootendorst, 2022; Mutlu et al., 2023).

This cumulative advantage mechanism creates self-reinforcing accessibility inequalities over time. Regions characterized by stronger healthcare infrastructures continue attracting additional healthcare investments, while underserved regions remain trapped within structurally weak accessibility environments. Consequently, market-driven healthcare distribution systems may unintentionally deepen existing social and spatial inequalities and further widen accessibility gaps between advantaged and disadvantaged communities.

Recent healthcare planning literature increasingly frames pharmacy accessibility within the broader context of the Social Determinants of Health (SDOH) framework. According to this perspective, health outcomes are shaped not only by healthcare services themselves but also by wider socioeconomic conditions including income, education, employment, housing quality, transportation access, neighborhood characteristics, and social environments. Consequently, pharmaceutical accessibility should be viewed as one component of a larger ecosystem of social determinants that collectively influence healthcare outcomes and population well-being (Chen et al., 2024; Adepoju et al., 2023).

Within this framework, an important distinction emerges between healthcare equality and healthcare equity. Equality-oriented approaches seek to provide similar levels of healthcare resources to all regions, whereas equity-oriented approaches recognize that different communities possess different healthcare needs and therefore may require different levels of accessibility protection. Communities characterized by higher levels of socioeconomic deprivation, population aging, chronic disease prevalence, or transportation disadvantage may require additional healthcare resources in order to achieve equitable healthcare outcomes. Consequently, modern healthcare planning increasingly emphasizes equity-sensitive allocation mechanisms rather than purely equal distribution policies (Han et al., 2025; Delamater, 2013).

Transportation poverty constitutes another critical socioeconomic determinant affecting pharmaceutical accessibility. Transportation poverty refers to situations in which individuals lack sufficient mobility resources to access essential services efficiently. Households without private vehicles, populations residing in transit-poor

environments, and individuals facing financial transportation barriers frequently experience reduced healthcare accessibility even when healthcare facilities are located within relatively short geographic distances. Transportation inequality therefore functions as a hidden mechanism reinforcing healthcare disparities across urban systems (Mutlu et al., 2023; Chen et al., 2024).

The influence of demographic aging on pharmaceutical demand has become increasingly significant in both developed and developing healthcare systems. Aging populations generally exhibit higher prevalence rates of chronic diseases, polypharmacy requirements, long-term treatment dependence, and healthcare utilization intensity. Consequently, elderly populations generate substantially higher pharmaceutical demand compared with younger demographic groups. Studies consistently demonstrate that accessibility barriers affecting elderly populations produce disproportionately large impacts on healthcare outcomes because these groups are more dependent on continuous medication use and routine healthcare interactions (Pednekar & Peterson, 2018; Han et al., 2025).

Urban geography literature further highlights the role of spatial segregation in shaping healthcare accessibility inequalities. Residential segregation based on income, education, housing markets, or social stratification frequently results in uneven distributions of healthcare infrastructure. Affluent districts generally attract greater healthcare investment and service diversity, whereas disadvantaged neighborhoods often experience lower service density and reduced accessibility. Over time, these patterns create cumulative advantage mechanisms through which existing inequalities become increasingly reinforced and institutionalized (Grootendorst, 2022; Catalano et al., 2024).

The theory of cumulative disadvantage provides an additional explanation for the persistence of healthcare accessibility inequalities. According to this perspective, social disadvantages rarely operate independently. Instead, multiple vulnerability factors interact and reinforce one another over time. Low income, weak transportation accessibility, advanced age, poor healthcare literacy, and limited digital skills frequently coexist within the same communities. Consequently, the combined effect of these disadvantages often exceeds the sum of their individual impacts, creating severe barriers to healthcare access and service utilization (Han et al., 2025; Chen et al., 2024).

Recognizing these interactions, recent facility location and healthcare optimization studies increasingly incorporate vulnerability-sensitive indicators directly into analytical models. Rather than relying solely on population counts or average travel distances, contemporary healthcare planning frameworks frequently utilize weighted demand structures that account for demographic, socioeconomic, and accessibility-related characteristics simultaneously. Such approaches allow optimization models to better represent real-world healthcare needs and improve the fairness of resource allocation decisions (Daskin, 2013; Murray, 2015; Özceylan et al., 2021).

In the Turkish context, similar socioeconomic accessibility dynamics have increasingly been documented within healthcare accessibility and urban planning studies. Although population-based regulatory mechanisms aim to ensure balanced pharmaceutical service distribution, they often fail to adequately capture income heterogeneity, mobility constraints, transportation inequalities, healthcare vulnerability differences, and differentiated regional healthcare needs (Özdemir, 2020; Yılmaz & Gökçe, 2021). Therefore, compliance with population-based regulations does not necessarily guarantee equitable pharmaceutical accessibility across urban systems.

In semi-urban and developing province centers, pharmacies frequently cluster around commercially active districts, shopping corridors, and central urban zones characterized by strong pedestrian mobility and intense economic activity. Conversely, newly urbanized peripheral settlements and socioeconomically disadvantaged neighborhoods may continue experiencing inadequate healthcare accessibility despite population growth and increasing healthcare demand. These patterns reveal the limitations of purely demographic regulatory structures in addressing multidimensional healthcare accessibility inequalities.

Another important issue highlighted in recent literature concerns the nonlinear interaction among socioeconomic determinants. Healthcare accessibility inequalities rarely emerge from a single variable alone. Instead, multiple vulnerability factors interact simultaneously and intensify one another. For example, low income amplifies the effects of weak transportation infrastructure, while aging intensifies the consequences of spatial isolation and limited healthcare mobility. Similarly, low healthcare literacy combined with digital exclusion may significantly worsen healthcare accessibility conditions even in relatively well-served urban environments.

Ignoring these nonlinear interactions risks generating structurally biased healthcare planning outcomes. Optimization models or regulatory systems relying solely on population density or average-distance metrics may unintentionally prioritize already advantaged regions while underrepresenting the needs of vulnerable populations. Consequently, multidimensional vulnerability-sensitive planning frameworks are increasingly emphasized within healthcare optimization literature. These findings indicate that population size alone constitutes an insufficient proxy for healthcare need and accessibility conditions. To accurately represent healthcare accessibility inequalities, multidimensional vulnerability indices and weighted accessibility structures are required (Han et al., 2025; Catalano et al., 2024).

These developments demonstrate that socioeconomic determinants should not be treated merely as contextual background variables but rather as fundamental components of healthcare accessibility planning. Effective pharmacy distribution strategies therefore require integrated planning frameworks capable of simultaneously considering demographic demand, socioeconomic vulnerability, transportation accessibility, healthcare infrastructure availability, and spatial equity objectives. Such considerations provide the theoretical basis for incorporating weighted demand structures and equity-oriented objectives within healthcare facility location models.

In the present study, these socioeconomic determinants are embedded directly into the optimization framework developed for Ağrı city center. Rather than treating vulnerability as an expost evaluation outcome, socioeconomic and accessibility-related indicators are integrated into the decision-making structure as central model inputs. Through weighted demand structures, vulnerability-sensitive accessibility considerations, realistic service constraints, and equity-oriented objective functions, the proposed framework generates pharmacy distribution patterns that more accurately reflect real-world healthcare heterogeneity. By incorporating socioeconomic vulnerability directly into the optimization process, the model seeks not only to improve overall accessibility performance but also to reduce structural healthcare inequalities and support more equitable pharmaceutical service provision across the urban system.

2.4. Regulatory Framework and Spatial Inequalities in Türkiye

The spatial configuration of community pharmacies in Türkiye is shaped not only by market dynamics but also by regulatory mechanisms that directly influence the establishment, relocation, and operation of pharmacies. The most influential of these mechanisms is the population-based regulatory framework introduced through the Regulation on Pharmacists and Pharmacies in 2012. This framework links the establishment of new pharmacies to predefined population thresholds and seeks to maintain a balanced distribution of pharmaceutical services across different regions of the country. In principle, such a regulatory structure aims to prevent excessive pharmacy concentration in commercially attractive locations while simultaneously ensuring a minimum level of service coverage for the broader population.

The introduction of population-based regulation represented a significant policy intervention within the Turkish pharmaceutical sector. Prior to the implementation of this framework, pharmacy distribution was influenced more heavily by market conditions and individual investment decisions, which often resulted in substantial concentration within commercially attractive urban areas. By establishing demographic thresholds for new pharmacy openings, policymakers aimed to create a more controlled and equitable pharmaceutical service environment. The underlying assumption was that population size could serve as a practical proxy for healthcare demand and pharmaceutical service requirements.

However, empirical evidence increasingly suggests that population-based regulation remains insufficient for capturing the multidimensional nature of healthcare accessibility and spatial inequality (Özdemir, 2020; Yılmaz & Gökçe, 2021). Population thresholds often treat urban and regional spaces as homogeneous units despite substantial variations in demographic composition, socioeconomic conditions, transportation accessibility, healthcare vulnerability, and service demand characteristics. As a result, the regulatory framework may create a superficial appearance of balance while concealing localized accessibility deficits at the neighborhood level.

At a conceptual level, population-based regulation assumes that population size is a reliable predictor of healthcare demand and pharmaceutical accessibility needs. Nevertheless, contemporary healthcare accessibility research increasingly challenges this assumption. Population counts alone cannot adequately represent variations in age

structure, chronic disease prevalence, healthcare dependency, mobility limitations, transportation availability, income inequality, healthcare literacy, or socioeconomic vulnerability. Consequently, regions with identical population sizes may exhibit substantially different healthcare accessibility conditions and service requirements.

For example, two districts with the same population may differ significantly in terms of age composition, healthcare utilization intensity, transportation infrastructure, and socioeconomic conditions. One district may consist primarily of younger and highly mobile residents, while another may contain a large proportion of elderly individuals, chronically ill patients, and low-income households. Despite these differences, both districts may be evaluated similarly under a rigid population-based regulatory framework (Özdemir, 2020). This limitation illustrates how demographic thresholds may overlook important qualitative dimensions of healthcare demand.

Socio-Spatial Cleavages and Spatial Pull

The limitations of population-based regulation become particularly visible in large metropolitan regions and developing provincial centers characterized by heterogeneous urban structures and uneven socioeconomic development patterns. Within such environments, pharmacies naturally tend to cluster around commercially attractive districts characterized by high pedestrian mobility, strong purchasing power, dense retail activity, and favorable economic conditions. These areas provide greater operational security, stronger customer bases, and more predictable financial performance, making them attractive locations for pharmacy investment.

In contrast, peripheral urban districts, newly urbanized settlements, rural communities, and socioeconomically disadvantaged neighborhoods frequently remain underserved despite often exhibiting greater healthcare vulnerability and higher healthcare needs (Aliağaoğlu & Uğur, 2013; Uğur, 2017). This phenomenon creates a distinct socio-spatial cleavage within urban systems. Market-oriented location preferences direct investment toward existing infrastructure nodes and economically dynamic districts, while geographically isolated or disadvantaged communities receive comparatively less attention.

Empirical studies conducted within Türkiye provide strong support for these observations. Aliağaoğlu and Uğur (2013), in their spatial analysis of pharmacies in Ankara, identified significant clustering patterns in central districts accompanied by

declining accessibility levels toward peripheral urban areas. Uğur (2017) expanded this perspective to the national scale and revealed persistent regional inequalities in pharmacy distribution throughout Türkiye. These findings indicate that formal regulatory compliance does not automatically guarantee equitable healthcare accessibility outcomes.

In other words, population-based legal restrictions may be effective in limiting excessive concentration, yet they do not actively guide pharmaceutical investments toward underserved regions. Regulatory compliance and healthcare equity therefore represent related but fundamentally different objectives. A region may satisfy demographic regulations while simultaneously exhibiting substantial accessibility inequalities at finer spatial scales.

International Perspectives on Pharmacy Regulation

The challenges associated with population-based pharmacy regulation are not unique to Türkiye. Similar regulatory approaches have historically been implemented across various European and international healthcare systems with the objective of controlling pharmacy growth and maintaining minimum service standards. However, international evidence increasingly demonstrates that demographic threshold-based regulations often struggle to respond effectively to heterogeneous local accessibility conditions and rapidly changing urban environments (Murray, 2015; Daskin, 2013).

Many healthcare systems have gradually moved beyond purely demographic planning approaches and increasingly incorporate accessibility-oriented criteria into healthcare facility planning. These criteria frequently include travel time, transportation accessibility, service coverage, healthcare vulnerability, socioeconomic deprivation, and spatial equity indicators. Such developments reflect a growing recognition that healthcare accessibility cannot be adequately represented through population counts alone.

An important distinction emerging from international healthcare planning literature concerns the difference between regulatory efficiency and healthcare equity. Population-based regulations are generally designed to achieve administrative simplicity, controlled market growth, and institutional stability. While these objectives remain important, they do not necessarily ensure equitable healthcare accessibility outcomes. A regulatory system may successfully limit excessive facility concentration

while simultaneously failing to improve accessibility conditions for vulnerable populations residing in underserved areas.

Consequently, healthcare planning scholars increasingly argue that regulatory systems and optimization-based planning tools should be viewed as complementary rather than competing mechanisms. Regulations establish legal boundaries and minimum standards, whereas analytical models provide systematic tools for identifying efficient and equitable facility allocation patterns within those boundaries. The integration of regulatory structures with evidence-based optimization methods therefore represents a promising direction for contemporary healthcare planning.

Temporal Rigidity and Vulnerability Blind Spots

The Turkish pharmacy regulatory framework has contributed significantly to controlling uncontrolled pharmacy expansion in certain urban regions. Nevertheless, its reliance on static demographic thresholds limits its ability to respond effectively to rapidly changing urban and regional conditions. Cities and healthcare systems continuously evolve due to migration, demographic aging, transportation investments, economic restructuring, urban expansion, and changing healthcare demand patterns.

Despite these dynamic conditions, population thresholds often remain fixed for extended periods. This temporal rigidity may create substantial mismatches between actual healthcare needs and formal eligibility criteria for pharmacy establishment (Özdemir, 2020; Yılmaz & Gökçe, 2021). Rapid population growth, changing settlement patterns, or emerging healthcare demand concentrations may therefore remain insufficiently addressed within existing regulatory frameworks.

Another important limitation concerns the absence of vulnerability-sensitive planning criteria. Current regulations do not explicitly differentiate among populations according to healthcare vulnerability, chronic disease prevalence, age structure, transportation disadvantage, mobility restrictions, or socioeconomic deprivation. Consequently, neighborhoods characterized by high levels of healthcare need may receive the same regulatory treatment as relatively advantaged communities.

This issue becomes increasingly significant in the context of population aging and urban expansion. Elderly individuals and chronically ill patients generally require more frequent interactions with pharmaceutical services than younger populations. However, these groups are simultaneously more sensitive to transportation barriers

and accessibility limitations. As a result, regulatory systems based solely on population density may underestimate pharmaceutical service requirements in vulnerable communities.

From Descriptive Analytics to Prescriptive Optimization

Descriptive GIS-based studies have played an important role in revealing spatial inequalities and accessibility imbalances within pharmaceutical service systems. By mapping pharmacy locations, service coverage areas, transportation accessibility patterns, population distributions, and demographic characteristics, GIS-supported analyses provide valuable visual diagnostics of healthcare accessibility disparities (Aliagaoglu & Uğur, 2013; Uğur, 2017). These studies have significantly improved understanding regarding how pharmaceutical services are distributed across urban environments and where accessibility gaps emerge.

The growing availability of spatial datasets, demographic information systems, and Geographic Information Systems (GIS) technologies has substantially enhanced the ability of researchers and policymakers to identify underserved areas. Through spatial visualization techniques, planners can observe service concentration patterns, evaluate travel distances, measure accessibility performance, and identify vulnerable communities that experience inadequate healthcare access. Such analyses have become increasingly important for understanding the spatial dimensions of healthcare inequality.

However, despite their analytical value, most GIS-based studies remain largely retrospective and descriptive in nature. They successfully identify where accessibility inequalities exist, but they generally do not provide structured guidance regarding how such inequalities should be corrected. Consequently, policymakers often remain limited to reactive interventions rather than proactive and optimization-driven planning strategies. In many cases, healthcare accessibility problems are recognized only after inequalities have already become entrenched within the urban system.

This distinction between descriptive and prescriptive planning represents one of the most important methodological challenges in contemporary healthcare accessibility research. Descriptive approaches answer questions such as where accessibility gaps exist, which populations are underserved, and how pharmaceutical services are currently distributed. Prescriptive approaches, on the other hand, seek to determine

where healthcare facilities should be located in order to improve accessibility outcomes and reduce inequalities. The latter requires analytical frameworks capable of evaluating alternative facility configurations and selecting the most effective allocation patterns.

Without prescriptive analytical tools, healthcare policy responses frequently remain fragmented and short-term oriented. Temporary subsidies, exceptional permits, administrative adjustments, local interventions, or case-specific solutions may partially alleviate accessibility problems but rarely address the underlying structural causes of healthcare inequality. Consequently, many accessibility challenges persist despite repeated policy interventions.

This limitation has motivated increasing interest in optimization-based healthcare planning methodologies. Operations Research approaches provide systematic mechanisms for evaluating alternative facility location scenarios under multiple and often conflicting objectives. Rather than relying solely on intuition, administrative judgment, or static regulations, optimization models enable planners to identify solutions that balance accessibility, equity, operational feasibility, and economic sustainability simultaneously (Murray, 2015; Özceylan et al., 2021).

Healthcare Equity Under Regulatory Constraints

A particularly important challenge within the Turkish regulatory environment concerns the balance between healthcare equity and operational sustainability. Community pharmacies operate within both healthcare and commercial systems. On one hand, they provide essential healthcare services and contribute directly to public health outcomes. On the other hand, they function as private enterprises that must maintain financial viability in order to continue operating effectively.

This dual structure creates inherent tensions between public health objectives and market-oriented incentives. Policies designed exclusively to maximize healthcare equity may impose financial burdens on pharmacy operators by encouraging investment in economically less attractive locations. Conversely, purely market-driven allocation mechanisms may intensify accessibility inequalities by concentrating pharmaceutical services within commercially advantageous districts.

Consequently, healthcare planners frequently face trade-offs among equity, efficiency, and financial sustainability. Improving accessibility for underserved

populations may require additional investments, while maximizing profitability may encourage clustering in already well-served areas. The challenge therefore lies not in prioritizing one objective at the expense of others but in identifying balanced solutions capable of simultaneously supporting accessibility, equity, and operational viability.

Recent healthcare planning literature increasingly argues that healthcare accessibility should be evaluated through multiple performance dimensions rather than a single criterion. These dimensions include average accessibility, worst-case accessibility, spatial equity, service continuity, healthcare resilience, workload balance, and operational sustainability. Such multidimensional evaluation frameworks provide a more comprehensive representation of healthcare system performance than traditional population-based planning approaches alone (Murray, 2015; Özceylan et al., 2021).

Why Optimization-Based Planning Is Necessary

From an analytical perspective, population-based regulations provide a necessary but insufficient basis for healthcare planning. While demographic thresholds can establish minimum service standards and control excessive facility growth, they cannot determine where facilities should be located to maximize accessibility performance under heterogeneous spatial conditions.

Optimization-based planning frameworks address this limitation by explicitly evaluating alternative location configurations and identifying solutions that balance multiple objectives simultaneously. Through mathematical programming techniques, planners can analyze how different allocation decisions influence accessibility, equity, service coverage, operational workload, infrastructure integration, and financial sustainability.

Unlike traditional planning approaches, optimization models provide the ability to evaluate thousands of alternative facility configurations simultaneously. This capability becomes particularly valuable in healthcare systems characterized by complex spatial relationships, heterogeneous demand structures, limited resources, and competing planning objectives. Rather than relying on subjective judgments, optimization models generate evidence-based recommendations grounded in measurable performance criteria.

Furthermore, optimization frameworks support scenario-based analysis under changing demographic, economic, and spatial conditions. This capability is especially important in rapidly evolving urban environments where healthcare demand patterns continuously change due to migration, population aging, transportation investments, urban expansion, and socioeconomic transformation. As a result, optimization models provide a level of adaptability and analytical flexibility that static demographic regulations cannot easily replicate.

The increasing complexity of contemporary healthcare systems therefore requires planning approaches capable of integrating accessibility, vulnerability, service continuity, operational feasibility, and healthcare equity considerations within a unified decision-support framework. Optimization methodologies provide precisely this capability and have consequently become central tools within modern healthcare planning literature.

Positioning the Proposed Framework

From a policy design perspective, the primary challenge is not to eliminate regulation but to support existing regulatory structures with analytical and optimization-based decision-support mechanisms. Such mechanisms can better represent real-world spatial heterogeneity, accessibility inequalities, and vulnerability differences than static demographic thresholds alone. A regulatory system supported by optimization-based healthcare planning tools can evolve from rigid rule enforcement toward adaptive, evidence-based, and accessibility-sensitive decision making.

Multi-objective optimization frameworks are particularly valuable because they allow healthcare planners to evaluate competing objectives simultaneously. Rather than focusing exclusively on average accessibility or service coverage, these frameworks can incorporate multiple performance dimensions, including worst-case accessibility, spatial equity, workload balancing, healthcare infrastructure integration, and long-term service sustainability. Consequently, optimization approaches provide more flexible and vulnerability-sensitive planning alternatives compared with traditional demographic regulations.

The integration of optimization methodologies into healthcare planning also supports greater transparency and accountability in decision making. Because

mathematical models explicitly define objectives, constraints, assumptions, and performance measures, policymakers can more clearly evaluate the consequences of alternative planning decisions. This transparency contributes to more objective and evidence-based healthcare planning processes.

In the present study, the Turkish regulatory environment is treated as an operational boundary condition rather than as the sole determinant of optimality. The proposed framework does not merely reproduce the outcomes generated by existing population-based regulations. Instead, it evaluates how accessibility, equity, service balance, healthcare infrastructure integration, and sustainability objectives can be improved while remaining consistent with the existing regulatory environment.

Within this context, the maximum regulatory capacity for the study area is represented by ($P=41$) pharmacies. This value establishes the upper planning boundary while allowing the optimization model to determine the most effective distribution pattern under multiple accessibility and equity considerations. In this manner, regulation functions as a practical constraint rather than a substitute for analytical decision making.

Through this perspective, the study enables scenario-based accessibility analyses under alternative pharmaceutical service configurations. By integrating deterministic proxy opening costs, weighted demand structures, physical accessibility constraints such as ($D_{max}=3000$) meters, healthcare infrastructure considerations, and GIS-supported spatial analysis into a unified optimization framework, the proposed model provides a more adaptive and analytically robust approach to pharmaceutical service planning in Türkiye.

Ultimately, the discussion presented in this section demonstrates that healthcare accessibility inequalities in Türkiye cannot be adequately addressed through static population-based regulations alone. While existing regulations provide important institutional safeguards and service standards, they remain limited in their ability to account for spatial heterogeneity, vulnerability differences, transportation inequalities, and evolving healthcare demand structures. Consequently, multidimensional, equity-sensitive, and optimization-supported planning approaches are increasingly necessary to ensure sustainable, balanced, and socially responsive pharmaceutical service distribution across heterogeneous urban and regional environments.

The proposed framework developed in this study contributes to this objective by combining regulatory awareness, healthcare accessibility principles, spatial analysis capabilities, and multi-objective optimization techniques within a single integrated decision-support structure. Through this integration, the framework seeks not only to evaluate existing accessibility conditions but also to proactively identify more equitable and sustainable pharmaceutical service configurations for future healthcare planning efforts.

2.5. Facility Location Theory and Classical Optimization Models

Facility location theory constitutes one of the most established and influential research areas within Operations Research, spatial analysis, logistics, and optimization. The field focuses on determining the optimal placement of facilities in order to serve geographically distributed demand while satisfying operational, economic, spatial, and policy-related constraints. Since its emergence in the mid-twentieth century, facility location theory has evolved into a sophisticated collection of mathematical models designed to support strategic decision-making across numerous sectors including healthcare, transportation, emergency response, telecommunications, manufacturing, retail distribution, and public infrastructure planning (Daskin, 2013; Murray, 2015).

At its most fundamental level, facility location theory addresses a deceptively simple question: where should facilities be located in order to serve spatially distributed demand most effectively? Despite its apparent simplicity, this question becomes highly complex when decision-makers must simultaneously consider limited resources, heterogeneous demand patterns, accessibility requirements, operational costs, service quality objectives, and equity concerns. Consequently, facility location problems have become one of the most important classes of optimization problems within modern decision science.

The significance of facility location decisions extends far beyond operational efficiency. Once facilities are established, their locations often remain fixed for extended periods and influence accessibility patterns, service quality, infrastructure utilization, transportation requirements, and economic outcomes for many years. Poor location decisions may generate persistent inefficiencies, accessibility inequalities, and unnecessary operational costs. Conversely, well-designed facility networks can

significantly improve resource utilization, service accessibility, and long-term system performance.

In healthcare planning, facility location decisions are particularly important because they directly affect public health outcomes. Unlike purely commercial distribution systems, healthcare facilities provide essential services that influence treatment accessibility, healthcare equity, disease management, emergency response capabilities, and overall population well-being. Consequently, healthcare facility location problems require balancing operational efficiency with broader social objectives such as accessibility, equity, and service continuity.

Historical Development of Facility Location Theory

The intellectual foundations of facility location theory can be traced to early studies in transportation planning, network design, and regional economics. However, the field began to develop into a rigorous mathematical discipline during the 1960s through the pioneering work of Hakimi (1964, 1965). These studies established the theoretical foundations that continue to underpin contemporary location-allocation research.

Prior to Hakimi's contributions, facility location decisions were often treated as heuristic planning problems without formal optimization structures. Decision-makers relied heavily on intuition, experience, and simplified spatial analyses. While such approaches occasionally produced acceptable results, they lacked systematic methods for evaluating alternative configurations and guaranteeing optimal solutions.

Hakimi fundamentally transformed this situation through the introduction of the vertex optimality theorem. This theorem demonstrated that for a broad class of network location problems, optimal facility locations could be found among the nodes of the network rather than anywhere within continuous space. This insight dramatically reduced computational complexity and enabled facility location problems to be formulated as discrete mathematical optimization models (Hakimi, 1964; Hakimi, 1965).

The significance of Hakimi's work extends beyond computational efficiency. By establishing a rigorous mathematical foundation for location decisions, his research transformed facility location planning from an intuitive managerial activity into a formal optimization discipline. Modern healthcare facility planning, logistics network

design, emergency service deployment, and public infrastructure allocation all build upon these foundational principles.

Following Hakimi's contributions, facility location theory expanded rapidly throughout the 1970s, 1980s, and subsequent decades. Researchers developed increasingly sophisticated models capable of addressing diverse planning objectives, including cost minimization, accessibility improvement, service coverage maximization, workload balancing, and equity enhancement. As computational capabilities improved, these models became applicable to increasingly large and complex real-world systems.

Network Location Problems and Location-Allocation Structures

Facility location problems are commonly classified as location-allocation problems because they involve two interconnected decisions. The first decision concerns facility placement: determining which candidate locations should host facilities. The second decision concerns demand allocation: determining how demand points should be assigned to selected facilities.

These decisions are inherently interdependent. The optimal assignment pattern depends on facility locations, while the desirability of a facility location depends on the demand assigned to it. Consequently, location-allocation problems require simultaneous optimization of both decisions.

Within healthcare systems, location-allocation models are particularly valuable because healthcare demand is rarely distributed uniformly across geographic space. Population density, demographic composition, transportation accessibility, healthcare vulnerability, and socioeconomic conditions vary substantially across regions. Location-allocation frameworks enable planners to explicitly incorporate these spatial differences into analytical decision-making processes.

The mathematical structure of location-allocation problems typically consists of demand nodes, candidate facility locations, distance or travel-time matrices, facility capacities, budget limitations, and assignment variables. Optimization models seek to identify facility configurations that maximize performance according to predefined planning objectives while satisfying operational constraints.

Because of their flexibility, location-allocation models have been applied extensively within healthcare planning. Hospitals, emergency medical service stations,

vaccination centers, pharmacies, blood banks, healthcare warehouses, and primary care facilities have all been analyzed using location-allocation methodologies. The widespread adoption of these models reflects their ability to convert complex healthcare planning problems into structured mathematical decision frameworks.

The P-Median Model: Efficiency-Oriented Planning

Among classical facility location formulations, the P-median model is arguably the most widely applied within healthcare accessibility research. Originally developed to address service allocation problems within network environments, the P-median model seeks to minimize the total weighted distance between demand points and assigned facilities.

The central objective of the P-median formulation is efficiency. By minimizing aggregate travel distance, the model reduces transportation costs, improves average accessibility conditions, and increases overall system performance (Daskin, 2013; Özceylan et al., 2021). In healthcare applications, this objective corresponds to reducing average patient travel burdens and improving the efficiency of healthcare service delivery.

A key strength of the P-median model lies in its ability to account for heterogeneous demand levels. Demand points associated with larger populations or greater service requirements receive proportionally greater influence within the objective function. Consequently, the model naturally prioritizes locations capable of serving larger numbers of users efficiently.

Because of this objective structure, P-median solutions frequently concentrate facilities near population centers and highly demanded regions. Such allocations often produce excellent average accessibility performance and efficient resource utilization. For this reason, the P-median model has been extensively applied in hospital planning, emergency service deployment, healthcare distribution systems, and pharmacy accessibility studies.

However, the efficiency orientation of the P-median model also introduces important limitations. By focusing on aggregate accessibility performance, the model may underrepresent geographically isolated populations. Peripheral settlements, sparsely populated areas, and vulnerable communities often contribute relatively little to the objective function and may therefore receive lower priority during optimization.

This tendency may generate accessibility inequalities despite excellent average system performance. A facility network optimized solely according to P-median principles may successfully minimize average travel distance while simultaneously leaving certain communities with substantially poorer access conditions. Such outcomes are particularly problematic within healthcare systems because disadvantaged populations frequently exhibit greater healthcare needs rather than lower ones.

Several healthcare accessibility studies have highlighted this limitation. Research consistently demonstrates that minimizing average travel distance does not automatically guarantee equitable healthcare access, particularly for populations located in peripheral, rural, or socioeconomically disadvantaged regions (Pednekar & Peterson, 2018; Catalano et al., 2024). Consequently, reliance on efficiency-oriented planning alone may contribute to the formation of healthcare accessibility disparities and pharmacy deserts.

These concerns motivated the development of alternative location models that explicitly consider accessibility inequalities and worst-case service conditions. Among these alternatives, the P-center model emerged as one of the most influential equity-oriented facility location formulations and represents the second major pillar of contemporary healthcare accessibility planning.

2.5.1. P-Center Model: Equity-Oriented Planning

In contrast to the efficiency-oriented philosophy of the P-median model, the P-center model adopts an explicitly equity-oriented perspective. Originally introduced by Hakimi (1965), the P-center problem seeks to minimize the maximum distance between any demand point and its nearest facility. Rather than focusing on average accessibility conditions, the model concentrates on protecting the most disadvantaged users within the system.

The fundamental logic of the P-center model is based on the minimax principle. Instead of minimizing total travel distance, the model minimizes the worst accessibility condition experienced by any demand point. Consequently, facility locations are selected to reduce the greatest accessibility burden within the service region. This objective aligns closely with public service planning philosophies emphasizing fairness, equity, and universal service provision (Murray, 2015).

Within healthcare systems, the P-center model offers several important advantages. Healthcare accessibility is often evaluated not only according to average service quality but also according to whether vulnerable populations can access essential services within acceptable travel distances. From this perspective, extremely poor accessibility conditions affecting even a small portion of the population may be considered unacceptable regardless of overall system efficiency.

By explicitly considering the worst-served demand points, the P-center model encourages facility placement closer to peripheral, rural, and underserved areas. Consequently, the model helps reduce spatial exclusion and improves accessibility conditions for geographically disadvantaged populations. This characteristic makes the P-center approach particularly attractive for healthcare planning applications involving hospitals, emergency medical services, vaccination centers, pharmacies, and other essential public services.

However, equity-oriented planning often involves important trade-offs. By prioritizing remote and sparsely populated areas, the P-center model may increase aggregate travel distances and reduce overall operational efficiency. Facilities may be positioned in locations that improve accessibility for isolated communities but serve fewer users overall. As a result, total transportation costs and infrastructure expenditures may increase relative to efficiency-oriented solutions.

Despite these limitations, the P-center model remains one of the most influential accessibility-based planning frameworks in healthcare facility location research. Its explicit focus on service equity provides an important counterbalance to efficiency-oriented models and has significantly influenced contemporary healthcare accessibility planning.

2.5.3. Coverage-Based Facility Location Models

Coverage-based models constitute another important family of facility location formulations. Unlike P-median and P-center approaches, which focus on distance minimization, coverage models evaluate whether demand points satisfy predefined accessibility standards.

The basic principle underlying coverage models is straightforward. A demand point is considered covered if it lies within a specified distance or travel-time threshold from a facility. Otherwise, it is classified as uncovered. This binary representation allows

planners to evaluate service coverage directly according to policy-defined accessibility standards.

One of the earliest and most influential coverage formulations is the Location Set Covering Problem (LSCP). The objective of LSCP is to identify the minimum number of facilities required to achieve complete coverage of all demand points (Daskin, 2013). This model is particularly useful when universal service provision represents the primary planning objective.

Another widely applied formulation is the Maximal Covering Location Problem (MCLP), introduced by Church and ReVelle (1974). Unlike LSCP, MCLP assumes that resources are limited and therefore seeks to maximize the covered population using a fixed number of facilities. This formulation has become especially popular in healthcare planning because healthcare resources are often constrained by budgetary, operational, and infrastructural limitations.

Coverage models have been extensively applied in ambulance deployment, emergency medical services, disaster response planning, fire station placement, and healthcare accessibility studies. Their popularity stems largely from their intuitive policy interpretation. Decision-makers can specify service standards such as a maximum travel time of 10 minutes or a maximum travel distance of 2 kilometers and directly evaluate whether populations satisfy these requirements.

Nevertheless, coverage models possess important limitations. Their binary structure may oversimplify complex accessibility conditions. Demand points located just inside and just outside a coverage threshold may exhibit nearly identical accessibility conditions while receiving completely different classifications. Similarly, demand points with substantially different travel distances may be treated identically if both satisfy the coverage threshold.

Consequently, coverage models often require supplementation with additional accessibility and equity measures when applied to complex healthcare planning environments.

Healthcare Facility Location Models

The application of facility location theory to healthcare planning has expanded considerably during the past several decades. Healthcare systems present unique planning challenges because they simultaneously involve accessibility objectives,

public welfare considerations, operational constraints, budget limitations, and equity requirements.

Unlike commercial facilities, healthcare facilities provide essential services that directly influence population health outcomes. Consequently, healthcare location decisions often carry broader social implications than traditional logistics or distribution problems. Poor healthcare accessibility may contribute to delayed treatment, lower healthcare utilization, reduced medication adherence, and increased healthcare inequalities.

Healthcare facility location models have therefore become important tools for planning hospitals, clinics, emergency medical services, vaccination centers, blood banks, pharmacies, and public health infrastructure. Numerous studies demonstrate that optimization-based healthcare planning can substantially improve service accessibility while simultaneously enhancing resource utilization and operational efficiency (Daskin, 2013; Murray, 2015).

Recent healthcare location studies increasingly emphasize the integration of accessibility and equity considerations into planning frameworks. Researchers have moved beyond purely cost-oriented models and now incorporate healthcare vulnerability, demographic heterogeneity, transportation accessibility, and social equity objectives into facility location decisions. This evolution reflects a broader transformation within healthcare planning toward more patient-centered and accessibility-sensitive decision-making approaches.

GIS and Healthcare Accessibility Planning

The development of Geographic Information Systems (GIS) has significantly expanded the analytical capabilities available to healthcare planners. GIS technologies allow researchers to visualize demand distributions, service coverage patterns, transportation networks, accessibility inequalities, and facility allocation structures within geographic space.

The integration of GIS with optimization models represents one of the most important developments in contemporary healthcare planning. While optimization models identify mathematically desirable facility configurations, GIS provides a spatial framework for interpreting and communicating these results. Together, these

tools enable decision-makers to evaluate accessibility conditions more comprehensively and transparently.

GIS-supported analyses are particularly valuable in healthcare planning because accessibility possesses an inherently spatial character. Distances, travel times, transportation infrastructure, topography, settlement patterns, and healthcare facility distributions all influence healthcare access. Consequently, GIS has become an indispensable component of modern healthcare accessibility research (Murray, 2015; Chen et al., 2024).

Hybrid and Multi-Objective Facility Location Models

One of the central challenges in healthcare facility planning concerns the trade-off between efficiency and equity. As discussed previously, P-median models prioritize efficiency while P-center models emphasize equity. Neither approach alone fully captures the complex requirements of contemporary healthcare systems.

To address this limitation, recent research increasingly adopts multi-objective optimization frameworks. These models integrate multiple planning objectives within a single mathematical structure, allowing decision-makers to evaluate trade-offs among accessibility, equity, operational feasibility, infrastructure integration, and economic sustainability simultaneously (Özceylan et al., 2021; Chen et al., 2024).

Among these approaches, hybrid P-median/P-center models have attracted substantial attention. These formulations simultaneously minimize average travel distance and maximum accessibility disparities. By integrating both objectives, hybrid models provide a more balanced representation of healthcare planning priorities than either approach individually.

Importantly, hybrid models do not eliminate the conflict between efficiency and equity. Instead, they explicitly quantify these trade-offs and allow decision-makers to adjust planning priorities according to local needs and policy objectives. This flexibility makes hybrid models particularly valuable in public healthcare planning environments characterized by multiple stakeholders and competing objectives.

Empirical studies consistently demonstrate that incorporating equity-oriented objectives into facility location models can substantially reduce accessibility disparities while maintaining acceptable levels of operational efficiency (Gu & Wang, 2019). Consequently, multi-objective optimization has emerged as one of the

dominant methodological paradigms within contemporary healthcare accessibility research.

Mixed Integer Linear Programming (MILP) Approaches

Many modern healthcare location-allocation problems are formulated as Mixed Integer Linear Programming (MILP) models. MILP provides a rigorous mathematical framework capable of representing binary facility location decisions, demand allocation relationships, capacity constraints, budget limitations, coverage requirements, and accessibility objectives simultaneously.

The popularity of MILP stems from its flexibility and ability to produce globally optimal solutions under clearly defined assumptions. Through binary and continuous decision variables, MILP models can capture highly complex healthcare planning environments while maintaining mathematical tractability.

Advances in optimization software and computational power have significantly expanded the practical applicability of MILP-based healthcare planning models. Contemporary solvers are capable of handling large-scale healthcare facility location problems involving thousands of decision variables and constraints, enabling increasingly realistic representations of healthcare systems.

As a result, MILP has become one of the most widely adopted optimization paradigms within healthcare facility location research, particularly in studies involving hospitals, pharmacies, emergency medical services, and healthcare accessibility planning.

Relevance to the Proposed Study

Although classical facility location models provide valuable theoretical foundations, real-world pharmacy planning requires additional considerations beyond travel distance alone. Community pharmacies operate simultaneously as healthcare service providers and economically sustainable private enterprises. Consequently, location decisions must balance accessibility objectives with operational feasibility, healthcare infrastructure integration, service continuity, and economic sustainability.

For this reason, the present study extends traditional facility location theory through the development of a multi-objective Mixed Integer Linear Programming framework specifically designed for pharmacy planning. The proposed model integrates classical

P-median and P-center formulations within a unified optimization structure while simultaneously incorporating additional planning dimensions.

In addition to accessibility and equity objectives, the framework incorporates deterministic proxy opening costs, healthcare facility proximity measures, weighted demand structures, service coverage considerations, and operational constraints. Through this integration, the model seeks to generate pharmacy distributions that more accurately reflect real-world healthcare planning requirements.

The proposed framework therefore represents a natural extension of the classical facility location literature. By combining efficiency-oriented accessibility planning, equity-sensitive service allocation, GIS-supported spatial analysis, healthcare infrastructure integration, and multi-objective optimization within a single decision-support structure, the model contributes to the growing body of research seeking to improve healthcare accessibility through advanced analytical methodologies.

Ultimately, the theoretical foundations presented in this section provide the conceptual basis for the hybrid P-median/P-center optimization framework developed in the subsequent methodology chapter. The proposed model builds directly upon these classical location theories while extending them to address the unique accessibility, equity, and operational challenges associated with pharmacy planning in Türkiye.

2.6. Multi-Objective Optimization and Equity-Oriented Planning

Healthcare planning problems are inherently multidimensional and involve multiple conflicting objectives that must be considered simultaneously. Decision-makers are typically required to balance operational efficiency, spatial accessibility, healthcare equity, service continuity, economic sustainability, infrastructure utilization, and long-term system resilience. Consequently, healthcare accessibility problems cannot be adequately represented through single-objective optimization frameworks that focus exclusively on one performance criterion while ignoring other important dimensions of system performance.

The complexity of healthcare systems stems from the fact that improvements in one objective frequently generate negative consequences for another. For example, minimizing average travel distance may improve overall accessibility efficiency but simultaneously worsen accessibility conditions for geographically isolated

populations. Similarly, maximizing service coverage may require substantial infrastructure investments that reduce financial sustainability. Therefore, healthcare planning inherently involves trade-offs among competing objectives, making multi-objective optimization an especially valuable analytical framework (Ehrgott, 2005; Murray, 2015).

Multi-objective optimization differs fundamentally from traditional single-objective approaches. Rather than searching for a single best solution according to one criterion, multi-objective models recognize the existence of competing goals and seek solutions that balance them appropriately. These approaches explicitly incorporate multiple performance measures into the decision-making structure and allow planners to evaluate how alternative solutions affect different stakeholders and planning priorities simultaneously.

Within healthcare systems, this capability is particularly important because accessibility and equity considerations often conflict with operational efficiency objectives. Public healthcare institutions are expected not only to maximize resource utilization but also to provide fair access to healthcare services for all citizens regardless of geographic location, socioeconomic status, age structure, or healthcare vulnerability. Consequently, optimization models that ignore equity considerations may produce technically efficient but socially undesirable outcomes.

The Evolution of Multi-Objective Thinking in Healthcare Planning

The growing importance of multi-objective optimization reflects a broader transformation within healthcare planning philosophy. Historically, healthcare systems were often evaluated primarily through efficiency indicators such as operational costs, facility utilization rates, or average service distances. While these measures remain important, contemporary healthcare planning increasingly recognizes that healthcare performance should also be evaluated through broader accessibility and equity perspectives.

This evolution has been influenced by advances in public health research, social epidemiology, healthcare geography, and spatial justice theory. Researchers increasingly acknowledge that average system performance may conceal substantial inequalities among population groups. Consequently, planning frameworks have gradually shifted from purely efficiency-oriented approaches toward models capable

of incorporating healthcare equity, vulnerability, and accessibility protection mechanisms.

The emergence of healthcare accessibility research further accelerated this transition. Numerous studies demonstrated that healthcare inequalities frequently persist even within systems characterized by relatively high average accessibility performance. These findings highlighted the limitations of relying exclusively on aggregate performance measures and motivated the development of more comprehensive planning methodologies.

Pareto Optimality and Multi-Objective Decision Making

A central concept within multi-objective optimization is Pareto optimality. A solution is considered Pareto optimal if no objective can be improved without simultaneously worsening at least one other objective. Rather than producing a single universally optimal solution, multi-objective optimization generates a set of efficient alternatives representing different trade-offs among competing objectives.

The collection of these solutions is commonly referred to as the Pareto frontier. Each point on the Pareto frontier represents a different balance among planning objectives. Decision-makers may then select the most appropriate solution according to policy priorities, stakeholder preferences, resource availability, and strategic goals.

Within healthcare planning, Pareto-based approaches are particularly valuable because healthcare systems rarely operate under a single objective. Accessibility, equity, operational feasibility, cost containment, healthcare resilience, and service quality must all be considered simultaneously. Consequently, Pareto-optimal frameworks provide planners with a transparent mechanism for evaluating competing planning priorities.

The Efficiency–Equity Trade-Off

One of the most extensively discussed issues within healthcare facility planning concerns the balance between efficiency and equity. This trade-off represents one of the fundamental challenges underlying healthcare accessibility optimizations.

Efficiency-oriented approaches generally seek to minimize total or average travel distance between demand points and healthcare facilities. These models improve aggregate system performance by reducing transportation costs, improving average

accessibility, and maximizing resource utilization. However, because they prioritize aggregate outcomes, such models frequently favor densely populated and economically active areas where larger numbers of individuals can be served efficiently.

As a result, peripheral communities, rural settlements, and socioeconomically vulnerable populations may receive lower priority within purely efficiency-oriented planning frameworks (Pednekar & Peterson, 2018; Catalano et al., 2024). While average accessibility may improve substantially, accessibility inequalities may simultaneously increase.

In contrast, equity-oriented planning focuses on improving accessibility conditions for underserved populations. Rather than prioritizing aggregate performance, equity-oriented models seek to reduce accessibility disparities and ensure minimum service standards throughout a region. The P-center model represents one of the most prominent examples of this philosophy because it explicitly minimizes the maximum accessibility burden experienced by any demand point (Murray, 2015).

From a healthcare accessibility perspective, equity-oriented approaches are especially valuable because they protect vulnerable populations from severe accessibility disadvantages. However, these improvements often come at the expense of increased operational costs, longer average travel distances, or reduced infrastructure utilization efficiency. Consequently, the conflict between efficiency and equity remains one of the central motivations for adopting multi-objective optimization frameworks.

Multi-Objective Healthcare Optimization

The inherent conflict between efficiency and equity has encouraged the development of hybrid and multi-objective optimization frameworks capable of simultaneously addressing multiple planning priorities. Rather than attempting to eliminate trade-offs, these models explicitly recognize their existence and provide systematic mechanisms for evaluating alternative compromises among competing objectives.

Multi-objective healthcare optimization offers several important advantages. First, it increases transparency by revealing the consequences of prioritizing one objective over another. Second, it supports scenario-based analysis by allowing planners to

evaluate alternative policy strategies under different assumptions and priorities. Third, it promotes balanced decision-making by ensuring that efficiency, accessibility, equity, and sustainability considerations are simultaneously represented within the planning framework.

Several empirical studies demonstrate the effectiveness of multi-objective healthcare planning approaches. Pourrezaie-Khaligh et al. (2022) showed that incorporating equity constraints significantly improves accessibility outcomes for vulnerable populations while maintaining acceptable levels of operational efficiency. Similarly, Salami et al. (2023) found that relatively small increases in operational costs can generate substantial improvements in accessibility equity and healthcare service distribution.

These findings indicate that healthcare planning should not rely exclusively on cost minimization or efficiency maximization. Accessibility improvements achieved through equity-sensitive optimization frequently generate broader societal benefits, including improved healthcare continuity, enhanced treatment adherence, reduced accessibility disparities, and greater healthcare resilience. Consequently, healthcare systems increasingly view accessibility and equity not as secondary considerations but as central components of long-term healthcare planning.

Equity Metrics in Healthcare Accessibility Planning

The growing emphasis on healthcare equity has led researchers to develop increasingly sophisticated methods for measuring accessibility disparities. Traditional planning approaches frequently relied on average accessibility indicators, which often concealed substantial inequalities among different population groups. In response, contemporary healthcare planning literature increasingly incorporates equity-sensitive performance measures.

Common equity indicators include maximum travel distance, accessibility variance, accessibility standard deviation, coverage inequality indices, service gap measures, and vulnerability-weighted accessibility metrics. These indicators enable planners to identify not only average system performance but also the distribution of accessibility outcomes across different communities.

In healthcare planning environments, equity metrics are particularly important because healthcare demand is inherently heterogeneous. Elderly populations, low-

income communities, chronically ill individuals, and geographically isolated settlements frequently require greater accessibility protection than the average population. Consequently, optimization frameworks that explicitly incorporate equity measures are generally more capable of addressing real-world healthcare accessibility challenges.

Recent studies further emphasize that healthcare accessibility should be evaluated not only through geographic distance but also through vulnerability-sensitive indicators. Population age structure, socioeconomic deprivation, mobility limitations, healthcare dependency, transportation accessibility, and chronic disease prevalence increasingly serve as important inputs within accessibility assessment frameworks. These developments reflect a broader shift toward patient-centered and socially responsive healthcare planning.

Healthcare Resilience and Sustainable Accessibility

Another important consideration within modern healthcare planning concerns healthcare resilience. Healthcare resilience refers to the ability of healthcare systems to maintain essential service provision during periods of disruption, uncertainty, or extraordinary demand.

The COVID-19 pandemic highlighted the critical importance of resilient healthcare infrastructure. During the pandemic, healthcare systems worldwide experienced substantial operational stress due to increased patient demand, workforce shortages, transportation disruptions, and supply chain interruptions. These experiences demonstrated that healthcare accessibility planning should consider not only routine operating conditions but also the ability of systems to function effectively during emergencies.

Multi-objective optimization frameworks contribute directly to healthcare resilience by supporting diversified service coverage, balanced facility allocation, accessibility redundancy, and improved infrastructure integration. Rather than concentrating services exclusively within highly efficient locations, equity-oriented planning approaches frequently distribute healthcare resources more evenly throughout geographic space. Such distributions often enhance resilience by reducing dependency on a small number of highly centralized service nodes.

Within pharmaceutical service systems, resilience considerations are particularly important because pharmacies often represent the most accessible healthcare facilities during emergencies. Consequently, resilient pharmacy networks contribute significantly to maintaining healthcare continuity during periods of disruption.

Weighted Objective Functions and Hybrid Optimization Structures

One of the most common approaches for implementing multi-objective optimization involves weighted objective functions. Under this approach, multiple objectives are combined into a single mathematical formulation through weighting coefficients that represent the relative importance of different planning priorities.

Weighted formulations provide substantial flexibility because they allow planners to adjust optimization behavior according to local conditions and policy preferences. Increasing the weight assigned to accessibility objectives, for example, may generate more equitable service distributions. Conversely, increasing efficiency-related weights may improve overall system performance while potentially reducing equity outcomes.

Within healthcare facility planning, weighted objective structures have become increasingly common because healthcare systems rarely operate under a single objective. Accessibility, equity, operational sustainability, infrastructure integration, service continuity, and economic feasibility must all be considered simultaneously. Weighted optimization frameworks provide a practical mechanism for balancing these competing requirements.

Furthermore, weighted objective functions facilitate sensitivity analysis and scenario evaluation. Decision-makers can systematically examine how different weighting structures influence facility allocation outcomes, accessibility patterns, and service equity. Such analyses provide valuable insights into the trade-offs associated with alternative planning strategies.

Sparse MILP-Based Multi-Objective Models

Mixed Integer Linear Programming (MILP) has emerged as one of the most widely adopted methodologies for implementing multi-objective healthcare optimization frameworks. MILP models provide sufficient flexibility to represent facility location decisions, demand allocation structures, capacity restrictions, accessibility constraints, and multiple planning objectives within a unified mathematical framework.

Despite these advantages, large-scale healthcare location-allocation problems often generate substantial computational complexity. As the number of demand points, candidate locations, and assignment relationships increases, the number of decision variables may grow exponentially. Such growth can significantly increase computational requirements and solution times.

To address this challenge, researchers increasingly employ sparse optimization structures. Sparse formulations reduce computational complexity by eliminating infeasible or unnecessary decision variables before optimization begins. One common approach involves applying predefined accessibility thresholds to exclude demand-facility assignments that exceed acceptable service distances.

By restricting assignment possibilities to realistic accessibility ranges, sparse optimization structures can dramatically reduce model size while preserving solution quality. Consequently, computational resources are focused on meaningful decision alternatives rather than mathematically possible but practically irrelevant assignments.

Recent studies demonstrate that sparse MILP formulations can substantially improve computational efficiency without sacrificing optimality. These characteristics make sparse optimization particularly attractive for large-scale healthcare accessibility problems involving extensive demand networks and numerous candidate facility locations.

Relevance to the Proposed Framework

The concepts discussed throughout this section provide the theoretical foundation for the optimization framework developed in the present study. The pharmacy location-allocation problem addressed in this research involves multiple planning objectives that cannot be adequately represented through a single-objective formulation.

Specifically, the proposed framework simultaneously considers accessibility efficiency, spatial equity, healthcare infrastructure integration, and operational sustainability. To address these requirements, a hybrid multi-objective optimization structure is developed that combines classical P-median and P-center principles within a unified Mixed Integer Linear Programming framework.

The P-median component seeks to improve overall accessibility performance by minimizing weighted average travel distances across demand regions. Simultaneously,

the P-center component seeks to protect vulnerable and geographically disadvantaged populations by minimizing maximum accessibility burdens. Through this integration, the framework attempts to balance efficiency and equity rather than prioritizing either objective exclusively.

Additionally, the model incorporates weighted demand structures representing heterogeneous healthcare needs across neighborhoods, healthcare facility proximity considerations, realistic accessibility constraints, and spatial feasibility requirements. These extensions enable the framework to better represent the complex realities of pharmacy planning within heterogeneous urban environments.

To ensure computational tractability, the study employs a sparse MILP structure that eliminates infeasible assignment relationships exceeding the maximum accepted service distance. This approach significantly reduces computational complexity while preserving the mathematical integrity of the optimization process.

Ultimately, the proposed framework reflects the broader evolution of healthcare planning from single-objective, efficiency-oriented approaches toward multi-objective, equity-sensitive, and accessibility-centered decision-support systems. By integrating efficiency, equity, accessibility, and operational considerations within a unified optimization framework, the study contributes to the growing body of literature seeking to improve healthcare service distribution through advanced analytical methodologies.

The concepts presented in this section therefore serve as the direct theoretical basis for the mathematical formulation and optimization procedures developed in the subsequent methodology chapter. Through the integration of multi-objective optimization, equity-oriented planning principles, and sparse MILP structures, the proposed model seeks to provide a robust and practically implementable framework for pharmacy location planning in Türkiye.

2.7. Criteria Weighting and Integrated Decision Structures

One of the most important challenges in multi-objective optimization is determining the relative importance of competing planning criteria. While optimization models provide mathematical mechanisms for evaluating alternative solutions, the quality and policy relevance of these solutions depend heavily on how different objectives are prioritized within the decision-making process. Consequently,

criteria weighting represents a fundamental component of multi-objective healthcare planning because it directly influences how resources are allocated, how accessibility inequalities are addressed, and how planning priorities are translated into operational decisions.

In many optimization applications, weighting coefficients are assigned through heuristic assumptions, expert judgment, or simplified preference structures. Although these approaches are relatively easy to implement, they frequently introduce subjectivity and may reduce transparency and reproducibility (Greco et al., 2016; Ehrgott, 2005). Different stakeholders often possess different priorities, making it difficult to justify why a specific criterion should receive greater importance than another. As a result, the weighting process itself becomes a critical decision problem rather than merely a technical parameter selection exercise.

Within healthcare accessibility planning, weighting decisions carry particularly significant implications because they influence how healthcare resources are distributed among populations and geographic regions. Small changes in weighting structures may substantially alter optimization outcomes, affecting which communities receive greater accessibility protection and which areas become prioritized for healthcare investment. Consequently, weighting procedures should be systematic, transparent, and theoretically justified rather than arbitrarily determined.

The Importance of Criteria Weighting in Healthcare Planning

Healthcare systems operate within highly complex environments characterized by limited resources, heterogeneous populations, diverse healthcare needs, and competing policy objectives. Decision-makers are frequently required to balance efficiency, accessibility, equity, service continuity, operational feasibility, infrastructure integration, and financial sustainability simultaneously.

Under such conditions, it is impossible to optimize all objectives equally. Prioritizing one objective inevitably influences the performance of others. Therefore, criteria weighting serves as the mechanism through which planning priorities are translated into mathematical structures.

For example, assigning greater importance to accessibility-related criteria may improve service availability for underserved communities but may also increase operational costs. Conversely, prioritizing efficiency objectives may reduce average

travel distances while simultaneously creating accessibility disadvantages for vulnerable populations. Consequently, weighting structures should be interpreted not merely as mathematical coefficients but as explicit representations of healthcare policy priorities.

The significance of weighting becomes even more pronounced in healthcare systems because accessibility inequalities often have direct consequences for public health outcomes. Decisions regarding weighting therefore influence not only operational performance but also healthcare equity, treatment continuity, and social welfare.

Interdependence of Healthcare Accessibility Criteria

Healthcare accessibility is influenced by a wide range of interconnected factors, including:

- Population density
- Age structure
- Socioeconomic status
- Transportation accessibility
- Healthcare infrastructure availability
- Mobility limitations
- Chronic disease prevalence
- Healthcare dependency
- Urban development patterns
- Service continuity requirements

These variables rarely operate independently. Instead, they interact continuously and influence one another through complex causal relationships.

For example, elderly populations frequently coincide with regions characterized by weaker transportation accessibility and greater healthcare dependency. Similarly, low-income neighborhoods often experience both infrastructure deficits and reduced mobility opportunities. Areas with limited healthcare infrastructure may also exhibit lower healthcare utilization rates, which in turn influence future healthcare investment decisions.

Consequently, healthcare vulnerability emerges from the interaction of multiple factors rather than from any single variable alone. Accessibility inequalities are therefore multidimensional phenomena generated by overlapping demographic, socioeconomic, geographic, and infrastructural conditions.

This complexity creates significant challenges for healthcare planning. Models that evaluate criteria independently may fail to capture important interactions among variables and therefore produce incomplete representations of healthcare accessibility conditions. As a result, contemporary healthcare planning increasingly emphasizes integrated decision-making frameworks capable of evaluating interdependent criteria simultaneously.

Multi-Criteria Decision Making (MCDM) in Healthcare Planning

The growing complexity of healthcare planning has contributed to the widespread adoption of Multi-Criteria Decision Making (MCDM) methodologies. MCDM approaches provide structured frameworks for evaluating alternatives under multiple and frequently conflicting criteria.

Unlike traditional optimization approaches that focus on a single objective, MCDM methodologies explicitly recognize the existence of multiple planning priorities. These methods allow decision-makers to incorporate both quantitative and qualitative information into the evaluation process while maintaining transparency and analytical consistency.

Within healthcare planning, MCDM approaches have been applied extensively to hospital location selection, healthcare accessibility evaluation, emergency service planning, healthcare technology assessment, pharmaceutical service allocation, and public health policy development.

The increasing popularity of MCDM reflects the recognition that healthcare decisions rarely depend on a single factor. Instead, successful healthcare planning requires balancing multiple dimensions of performance simultaneously.

Analytical Hierarchy Process (AHP)

One of the most widely used MCDM techniques is the Analytical Hierarchy Process (AHP), developed by Saaty. AHP structures decision problems hierarchically and determines criterion weights through pairwise comparisons.

The primary advantage of AHP lies in its simplicity and interpretability. Decision-makers can systematically compare criteria two at a time and derive relative importance scores. Consequently, AHP has become one of the most commonly applied weighting methodologies in healthcare planning studies.

However, AHP assumes that criteria are independent from one another. This assumption may be unrealistic in healthcare accessibility planning because demographic, socioeconomic, and spatial variables frequently exhibit strong interactions. Therefore, although AHP remains useful, more advanced approaches are often required when interdependencies among criteria become significant.

Analytical Network Process (ANP)

To overcome the independence assumption of AHP, Saaty later developed the Analytical Network Process (ANP). ANP extends hierarchical decision structures by explicitly incorporating feedback relationships and interdependencies among criteria.

Within ANP, criteria can influence one another directly and indirectly. This feature is particularly valuable for healthcare accessibility planning because accessibility variables frequently interact through complex causal mechanisms.

For example, transportation accessibility may affect healthcare utilization, while healthcare infrastructure availability may influence transportation demand. Similarly, socioeconomic conditions may affect both mobility patterns and healthcare dependency simultaneously. ANP provides a framework capable of capturing such relationships.

As a result, ANP has become increasingly popular in healthcare facility location studies involving multiple interacting decision criteria.

DEMATEL and Causal Relationship Analysis

Decision-Making Trial and Evaluation Laboratory (DEMATEL) represents another important methodology for analyzing complex decision environments. Unlike traditional weighting approaches, DEMATEL focuses explicitly on identifying causal relationships among criteria.

The method enables researchers to distinguish between cause factors and effect factors within a system. Consequently, DEMATEL provides insights not only into

criterion importance but also into the structural relationships that generate system behavior.

Within healthcare accessibility planning, DEMATEL is particularly useful because healthcare systems contain numerous interacting variables. Transportation accessibility influences healthcare utilization, socioeconomic conditions affect healthcare demand, and healthcare infrastructure availability shapes accessibility outcomes. DEMATEL allows these relationships to be analyzed systematically.

Furthermore, DEMATEL improves decision transparency by revealing how changes in one criterion may propagate throughout the broader planning system.

WINGS Methodology and Integrated Weighting Structures

Recent healthcare planning studies increasingly employ the Weighted Influence Non-linear Gauge System (WINGS) methodology. WINGS extends traditional weighting approaches by incorporating both criterion importance and interaction intensity within a unified analytical framework.

Unlike conventional methods that focus exclusively on criterion weights, WINGS evaluates how criteria influence one another throughout the decision system. This capability is particularly valuable in healthcare accessibility planning because healthcare inequalities rarely emerge from isolated variables. Instead, accessibility outcomes are generated through interactions among demographic, socioeconomic, geographic, and infrastructural conditions.

The integration of WINGS and DEMATEL methodologies provides a particularly powerful framework for healthcare planning. DEMATEL identifies causal relationships among criteria, while WINGS quantifies their relative influence and interaction strength. Together, these methods enable more realistic representations of healthcare accessibility systems than traditional weighting procedures.

Weight Sensitivity and Policy Interpretation

An important advantage of structured weighting methodologies is their ability to support sensitivity analysis. Sensitivity analysis evaluates how changes in criterion weights influence optimization outcomes.

This capability is especially important in healthcare planning because different stakeholders frequently possess different priorities. Public health authorities may

prioritize accessibility equity, local governments may emphasize service coverage, while private operators may focus on operational sustainability.

Through sensitivity analysis, decision-makers can examine how alternative policy priorities affect healthcare accessibility outcomes. Such analyses improve transparency and facilitate evidence-based policy discussions.

Furthermore, weighting coefficients should be interpreted as representations of planning priorities rather than purely mathematical parameters. Higher weights assigned to elderly populations, for example, indicate stronger commitments to healthcare continuity and accessibility protection. Similarly, greater emphasis on disadvantaged communities reflects a policy preference toward distributive justice and healthcare equity.

Normalization and Integrated Objective Structures

One of the most important technical challenges in multi-objective optimization concerns the integration of criteria measured in different units. Accessibility distances, operational costs, infrastructure proximity measures, and equity indicators often possess substantially different numerical scales.

Without normalization, variables with larger numerical values may dominate optimization outcomes regardless of their actual policy importance. Consequently, normalization procedures are required to eliminate scale effects and ensure meaningful comparisons among objectives.

Normalization transforms different criteria into dimensionless quantities that can be combined within a unified objective function. This process improves model stability, enhances interpretability, and allows weighting coefficients to reflect genuine planning priorities rather than mathematical scale differences.

Application to the Proposed Framework

In the proposed study, criteria weighting plays a central role in the development of the multi-objective optimization framework. The model simultaneously considers multiple planning objectives, including:

- Weighted aggregate travel distance
- Maximum accessibility distance
- Proxy facility opening costs

- Healthcare facility proximity

Because these objectives are measured using different units, normalization constants (N_1 , N_2 , N_3 and N_4) are employed to eliminate scale effects and create a dimensionally consistent optimization structure.

Following normalization, convex weighting coefficients (α , β , γ , and δ) are introduced to represent planning priorities and policy preferences. These coefficients determine the relative importance assigned to efficiency, equity, economic sustainability, and healthcare infrastructure integration within the optimization framework.

This approach provides a flexible and transparent decision-support mechanism capable of evaluating alternative accessibility scenarios under different policy assumptions. By adjusting weighting structures, decision-makers can systematically explore the trade-offs among accessibility, equity, cost, and healthcare integration objectives.

Ultimately, the integration of structured weighting methodologies, normalization procedures, and multi-objective optimization principles enables the proposed framework to represent healthcare accessibility as a multidimensional decision problem rather than a purely distance-based allocation exercise. Through this perspective, the study contributes to the growing body of healthcare planning literature emphasizing transparency, equity, and evidence-based decision-making in facility location optimization.

2.8. Research Gaps and Positioning of the Present Study

Despite the growing body of literature on healthcare accessibility, spatial inequality, healthcare geography, and facility location optimization, several important conceptual, methodological, and practical gaps remain unresolved. Although previous studies have contributed significantly to understanding healthcare accessibility patterns and identifying spatial service inequalities, the integration of equity-sensitive planning principles into operational healthcare decision-making frameworks remains limited. Furthermore, many existing studies focus on identifying accessibility problems rather than developing implementable optimization-based solutions capable of addressing them systematically.

Descriptive and Diagnostic Predominance

The first major gap concerns the predominance of descriptive and diagnostic approaches within healthcare accessibility research. Geographic Information Systems (GIS)-based mapping techniques are widely used to visualize spatial inequalities, accessibility disparities, pharmacy distributions, and healthcare service coverage patterns. These approaches provide valuable spatial diagnostics and significantly improve understanding of accessibility problems within urban and regional systems.

However, such analyses remain fundamentally retrospective in nature. They identify where inequalities exist but generally fail to provide structured analytical mechanisms capable of determining how these inequalities should be systematically corrected through operational decision-making processes (Aliagaoglu & Uğur, 2013; Uğur, 2017; Özdemir, 2020). Consequently, many studies successfully diagnose accessibility problems without offering prescriptive frameworks capable of generating evidence-based allocation decisions.

This limitation is particularly important in healthcare planning because identifying underserved regions does not automatically reveal where new facilities should be located or how healthcare resources should be redistributed. As a result, policymakers often remain dependent on administrative judgment and ad hoc interventions rather than systematic optimization-based planning methodologies.

Single-Objective Dominance

The second major gap relates to the dominance of single-objective optimization formulations within healthcare facility planning literature. Many optimization-based healthcare accessibility studies primarily focus on minimizing average travel distance, total transportation cost, or aggregate operational expenditures. While such formulations improve overall system efficiency, they frequently generate unintended spatial inequalities by favoring densely populated, economically attractive, and geographically central regions (Pednekar & Peterson, 2018; Catalano et al., 2024).

This tendency emerges because aggregate efficiency measures naturally prioritize areas where larger populations can be served with lower average costs. Consequently, peripheral settlements, rural communities, and socioeconomically disadvantaged neighborhoods may receive limited attention despite often exhibiting greater healthcare vulnerability.

The literature increasingly recognizes that healthcare planning requires balancing efficiency with broader accessibility and equity considerations. Nevertheless, many existing studies continue to emphasize a single performance criterion, thereby limiting their ability to address the multidimensional nature of healthcare accessibility.

Dichotomous Modeling Interpretation

The third important gap concerns the treatment of classical facility location formulations within optimization literature. Models such as P-median and P-center are frequently presented as alternative or competing approaches rather than complementary planning structures.

The P-median model emphasizes efficiency by minimizing weighted aggregate travel distance, whereas the P-center model prioritizes equity by minimizing maximum accessibility burdens (Hakimi, 1964; Hakimi, 1965). Although both models address important dimensions of healthcare accessibility, many studies adopt only one formulation and therefore capture only a partial representation of healthcare planning requirements.

This dichotomous interpretation obscures the potential advantages of integrated hybrid formulations capable of simultaneously balancing efficiency-oriented and equity-oriented objectives. Consequently, the literature still contains a limited number of healthcare planning frameworks that explicitly combine these two classical perspectives within a unified optimization structure.

Ex Post Vulnerability Representation

The fourth major gap concerns the representation of vulnerability within healthcare optimization frameworks. In much of the existing literature, vulnerability is treated primarily as an ex post evaluation outcome rather than as an integral component of the optimization structure itself.

Many studies assess whether disadvantaged populations remain underserved after optimization results have been generated. However, relatively few frameworks directly embed vulnerability-sensitive criteria into the mathematical decision-making process (Chen et al., 2024; Han et al., 2025).

As a result, accessibility inequalities are often identified only after optimization has been completed. This reactive perspective limits the ability of planning frameworks to

proactively protect vulnerable populations during the allocation process. Contemporary healthcare planning increasingly advocates vulnerability-sensitive optimization approaches that explicitly incorporate demographic, socioeconomic, and healthcare dependency characteristics into decision structures.

National Prescriptive Modeling Deficit

The fifth major research gap concerns the limited number of prescriptive and equity-oriented optimization studies within the Turkish context. Existing national studies predominantly focus on descriptive spatial analyses, regulatory evaluations, healthcare accessibility diagnostics, and pharmacy distribution assessments.

While these studies contribute significantly to identifying accessibility inequalities and spatial clustering patterns, they generally do not provide formal optimization-based decision-support frameworks capable of guiding equitable pharmacy planning under multidimensional accessibility conditions (Özceylan et al., 2021; Yılmaz & Gökçe, 2021).

Consequently, there remains a substantial need for analytical models capable of supporting pharmacy planning decisions within Türkiye while simultaneously incorporating accessibility, equity, operational feasibility, and regulatory considerations.

Disregard for Geographic Boundaries and Land-Use Constraints

The sixth major research gap involves the treatment of geographic space within facility location models. Many classical optimization formulations implicitly assume that urban environments represent continuous and unrestricted planning spaces.

In practice, however, cities contain numerous land-use restrictions and geographical barriers. Military zones, industrial areas, protected environmental regions, transportation corridors, topographical obstacles, and legally restricted areas frequently limit where healthcare facilities can be established.

Despite this reality, many healthcare location studies fail to incorporate such constraints explicitly. Consequently, mathematically optimal solutions may not always be operationally feasible or legally implementable. This disconnect between theoretical optimization and practical implementation remains a significant limitation within the literature.

2.8.7. Strategic–Operational Disconnection

The seventh major gap concerns the separation of strategic facility location decisions from operational service planning. Traditional location-allocation studies typically determine where facilities should be located without considering how those facilities will operate after implementation.

However, healthcare accessibility depends not only on facility placement but also on operational service structures. Duty scheduling, workforce allocation, emergency service availability, service continuity, and workload balancing all influence accessibility outcomes.

Within pharmacy systems, this issue becomes particularly important because accessibility conditions differ substantially between normal operating periods and emergency duty-service periods. Despite this reality, relatively few studies integrate strategic facility location decisions with operational pharmacy scheduling structures.

As a result, many optimization models evaluate only the strategic dimension of accessibility while neglecting the operational mechanisms that determine how healthcare services are actually delivered.

Healthcare Infrastructure Integration Gap

An additional limitation within the literature concerns the insufficient integration of healthcare infrastructure relationships into facility location models. Most healthcare accessibility studies focus primarily on distances between demand points and healthcare facilities while treating healthcare institutions as isolated service nodes.

In reality, pharmacies function as components of broader healthcare ecosystems. Their effectiveness is strongly influenced by their proximity to hospitals, family health centers, outpatient clinics, and other healthcare providers. Patients frequently access pharmaceutical services immediately before or after receiving care from these facilities.

Despite this functional interdependence, relatively few optimization studies explicitly incorporate healthcare infrastructure proximity into objective functions or allocation structures. Consequently, important opportunities for improving healthcare service integration remain underexplored.

Facility Cost Representation Gap

Another important gap concerns the representation of facility establishment costs within healthcare optimization frameworks. Many studies either ignore opening costs entirely or assume uniform establishment costs across all candidate locations.

Such assumptions may oversimplify real-world planning conditions. In practice, facility establishment costs vary according to location characteristics, accessibility conditions, infrastructure availability, commercial attractiveness, and urban centrality.

The omission of realistic cost structures may produce solutions that are mathematically attractive but economically unrealistic. Consequently, there remains a need for optimization frameworks capable of simultaneously considering accessibility, equity, and location-dependent cost structures.

GIS–Optimization Integration Gap

Although Geographic Information Systems have become increasingly common within healthcare accessibility research, GIS and optimization methodologies often remain only partially integrated.

Many studies use GIS for visualization and accessibility assessment while performing optimization separately. As a result, spatial information frequently functions as a descriptive support tool rather than as an integrated component of the decision-making framework.

Recent healthcare planning literature increasingly advocates tighter integration between GIS-supported spatial analysis and mathematical optimization. Such integration enables healthcare planners to combine geographic realism with analytical rigor, thereby improving both interpretability and practical applicability of planning outcomes (Murray, 2015; Chen et al., 2024).

Methodological Positioning of the Present Study

The present study seeks to address these conceptual, methodological, and practical gaps through the development of a hybrid, equity-sensitive, GIS-supported, and operationally integrated optimization framework implemented using the real urban structure of Ağrı city center.

Methodologically, the study is structured as a two-stage optimization framework.

In the first stage, a four-component normalized Mixed Integer Linear Programming (MILP) model simultaneously minimizes weighted aggregate accessibility distance,

controls maximum accessibility disparities through a P-center component, incorporates facility opening cost structures, and minimizes healthcare infrastructure proximity penalties. Furthermore, the model introduces a geographic preprocessing layer that explicitly excludes restricted urban zones, particularly the 12th Mechanized Infantry Brigade areas, ensuring that all candidate pharmacy locations remain operationally feasible and legally implementable.

Unlike many previous studies, vulnerability-sensitive accessibility considerations are incorporated directly into the optimization structure through weighted demand representations rather than being evaluated only after solution generation. Similarly, healthcare infrastructure integration is explicitly represented through proximity-based objective components.

In the second stage, the strategic location decisions generated by the first stage are integrated with a multi-criteria operational duty-pharmacy scheduling model. This operational layer incorporates demand coverage requirements, healthcare facility coverage constraints, minimum inter-pharmacy separation distances, and mandatory rest intervals to support service continuity and workforce sustainability.

Through this integrated design, the study bridges the gap between strategic facility location planning and operational healthcare service delivery. Consequently, the proposed framework contributes not only to the facility location literature but also to the emerging body of research seeking to integrate accessibility planning, healthcare operations, GIS-supported analysis, and equity-oriented optimization within a unified decision-support structure.

Ultimately, the study positions itself at the intersection of healthcare accessibility planning, facility location theory, multi-objective optimization, healthcare geography, and operational healthcare management. By addressing the limitations identified throughout the literature, the proposed framework seeks to provide a more comprehensive, realistic, and policy-relevant approach to pharmacy planning in Türkiye.

2.9. Summary of the Literature and Conceptual Framework

The literature reviewed throughout this chapter demonstrates that pharmacy location planning constitutes a highly complex and multidimensional healthcare decision problem influenced by spatial accessibility, demographic heterogeneity,

socioeconomic vulnerability, transportation infrastructure, healthcare service demand, regulatory structures, and operational sustainability considerations. Rather than representing a simple facility placement exercise, pharmacy planning emerges as a strategic public health challenge requiring the simultaneous consideration of efficiency, equity, accessibility, and long-term service continuity objectives.

A consistent finding throughout the literature is that healthcare accessibility inequalities are structurally embedded within urban and regional systems rather than occurring randomly. Numerous studies indicate that elderly populations, low-income households, rural communities, socioeconomically disadvantaged neighborhoods, disabled individuals, and populations with limited mobility experience significantly greater barriers to healthcare access compared with more advantaged groups (Pednekar & Peterson, 2018; Han et al., 2025). These inequalities are further intensified by transportation limitations, uneven infrastructure development, healthcare dependency, and spatial segregation processes that accumulate over time.

The reviewed literature also highlights the critical role of community pharmacies within contemporary healthcare systems. Pharmacies no longer function solely as medication dispensing facilities but increasingly operate as accessible primary healthcare access points providing preventive healthcare services, medication counseling, chronic disease management support, and healthcare guidance. Consequently, the spatial distribution of pharmacies directly influences healthcare accessibility, treatment continuity, medication adherence, and broader public health outcomes.

An important conclusion emerging from the literature concerns the limitations of traditional population-based planning frameworks. Although demographic regulations aim to standardize healthcare service provision and prevent uncontrolled facility concentration, they frequently fail to capture differences in healthcare vulnerability, transportation accessibility, mobility restrictions, healthcare dependency, and socioeconomic disadvantage (Özdemir, 2020; Yılmaz & Gökçe, 2021). As a result, formal regulatory compliance does not necessarily guarantee equitable healthcare accessibility outcomes.

The literature further demonstrates that healthcare accessibility should be interpreted as a multidimensional phenomenon extending beyond simple geographic proximity. Contemporary healthcare accessibility research increasingly incorporates

socioeconomic conditions, transportation infrastructure, age structure, healthcare vulnerability, and spatial equity considerations into analytical frameworks (Chen et al., 2024; Catalano et al., 2024). This broader perspective reflects a growing recognition that healthcare accessibility depends not only on where services are located but also on the characteristics of the populations attempting to access them.

Another important finding concerns the contribution of Geographic Information Systems (GIS) to healthcare accessibility research. GIS-based analyses have significantly improved the ability of researchers and policymakers to visualize accessibility inequalities, identify underserved populations, evaluate service coverage patterns, and assess spatial disparities in healthcare provision (Aliagaoglu & Uğur, 2013; Uğur, 2017). However, the literature simultaneously reveals that most GIS-based studies remain descriptive in nature and generally do not provide optimization-based decision-support mechanisms capable of correcting identified inequalities.

From a theoretical perspective, facility location theory provides the mathematical foundation for healthcare accessibility planning. Classical facility location models such as P-median and P-center remain among the most influential analytical frameworks within healthcare planning literature. The P-median model emphasizes efficiency by minimizing weighted average travel distances, whereas the P-center model prioritizes equity by minimizing worst-case accessibility conditions (Hakimi, 1964; Hakimi, 1965; Murray, 2015). Together, these models establish the conceptual foundations for balancing efficiency and accessibility within healthcare facility planning.

The literature also demonstrates that healthcare planning problems are inherently multi-objective in nature. Accessibility, equity, cost efficiency, operational sustainability, infrastructure integration, and healthcare resilience frequently represent competing objectives that cannot be adequately addressed through single-objective optimization approaches. Consequently, recent studies increasingly advocate the adoption of hybrid and multi-objective optimization frameworks capable of balancing multiple planning priorities simultaneously (Pourrezaie-Khaligh et al., 2022; Salami et al., 2023).

A further conclusion concerns the growing importance of vulnerability-sensitive planning approaches. Contemporary healthcare accessibility studies increasingly recognize that healthcare needs are distributed unevenly across geographic space and

that certain population groups require greater accessibility protection than others. Accordingly, vulnerability-sensitive weighting structures, equity-oriented objective functions, and integrated decision-support systems have become increasingly prominent within healthcare optimization research (Han et al., 2025; Chen et al., 2024).

The literature review additionally reveals several important methodological gaps. Existing studies frequently remain descriptive rather than prescriptive, emphasize single-objective optimization rather than balanced decision-making, treat vulnerability as an ex post evaluation criterion rather than an integrated planning component, neglect operational scheduling considerations, and rarely incorporate geographic feasibility restrictions directly into optimization structures. These limitations collectively motivate the development of more comprehensive healthcare planning frameworks capable of integrating accessibility, equity, operational feasibility, and spatial realism within a unified analytical structure.

Based on the theoretical and methodological foundations identified throughout the literature, the conceptual framework of the present study is constructed around a sequential and integrated planning architecture. The framework combines geospatial analysis, accessibility planning, multi-objective optimization, and operational healthcare management within a unified decision-support structure.

More specifically, the conceptual framework consists of five sequential components:

1. Geospatial Data Inputs
2. Geographic Spatial Filtering
3. Strategic Multi-Objective MILP Optimization
4. Strategic Facility Location Decisions
5. Operational Duty Pharmacy Scheduling

The first component involves the collection and preparation of geospatial, demographic, accessibility, and healthcare infrastructure data. This stage establishes the analytical foundation upon which subsequent optimization procedures are built. Demand points, candidate facility locations, healthcare infrastructure elements, transportation relationships, and accessibility measures are integrated into a unified spatial database.

The second component consists of a geographic spatial filtering layer. Unlike many traditional facility location studies, the proposed framework explicitly considers real-world land-use restrictions and geographic feasibility constraints. Restricted urban zones, including military areas and other non-developable regions, are removed during preprocessing to ensure that all candidate locations remain operationally feasible.

The third component represents the core analytical engine of the framework: a strategic four-component normalized Mixed Integer Linear Programming (MILP) model. This model simultaneously incorporates weighted aggregate accessibility distance, maximum accessibility protection, facility opening costs, and healthcare infrastructure proximity considerations within a unified multi-objective optimization structure. Through normalization and convex weighting procedures, the model balances efficiency, equity, economic feasibility, and healthcare integration objectives.

The fourth component generates strategic facility location decisions. At this stage, the optimization framework identifies the most appropriate pharmacy locations and corresponding demand assignments according to the selected planning priorities and accessibility objectives. The resulting allocation structure establishes the long-term accessibility blueprint for the study area.

The fifth component extends beyond traditional facility location analysis by incorporating an operational duty-pharmacy scheduling layer. Rather than treating accessibility exclusively as a strategic location problem, the framework recognizes that healthcare accessibility is also influenced by operational service availability. Consequently, emergency duty scheduling, workload balancing, coverage requirements, and service continuity considerations are integrated into the planning process.

The overall conceptual framework is illustrated in Figure 1.



The conceptual foundation of the study rests upon six fundamental principles:

Multidimensional Accessibility Dynamics	Vulnerability-Sensitive Prioritization
Convex Multi-Criteria Balancing	Pre-Emptive Geospatial Filtering
Integrated Two-Stage Execution	Transparent Decision Support

Figure 1. Proposed Two-Stage Pharmacy Location and Duty Scheduling Framework

The principle of multidimensional accessibility dynamics recognizes that healthcare accessibility is shaped simultaneously by demographic, socioeconomic, spatial, and infrastructural factors. Vulnerability-sensitive prioritization ensures that disadvantaged populations receive explicit consideration within the optimization process. Convex multi-criteria balancing allows competing objectives to be integrated within a normalized mathematical structure. Pre-emptive geospatial filtering improves practical feasibility by incorporating real-world geographic constraints. Integrated two-stage execution links strategic planning decisions with operational service

delivery. Finally, transparent decision support enhances the interpretability and policy relevance of optimization outcomes.

Taken together, these principles establish the theoretical and conceptual foundation for the proposed pharmacy planning framework. The resulting structure moves beyond traditional accessibility analyses by integrating facility location theory, healthcare accessibility research, GIS-supported spatial analysis, multi-objective optimization, vulnerability-sensitive planning, and operational healthcare management within a single coherent framework.

Accordingly, the following chapter presents the mathematical formulation of the proposed Mixed Integer Linear Programming (MILP) model, the hybrid four-component optimization structure, the normalization framework, the accessibility and equity constraints, the GIS-supported analytical process, and the integrated pharmacy location-allocation and duty-pharmacy scheduling methodology developed within the scope of this research.

3. PROBLEM DEFINITION AND FRAMEWORK

3.1. Problem Definition

The primary objective of this research is to develop an equity-oriented, prescriptive, and optimization-based decision-support framework for determining the most suitable locations for community pharmacies within Türkiye, specifically applied and empirically validated under the urban structure of Ağrı city center. The study operates under geographical, demographic, socioeconomic, regulatory, and operational constraints in order to improve healthcare accessibility, reduce spatial inequalities in pharmaceutical service distribution, prevent excessive clustering among pharmacies, and support the long-term sustainability of pharmaceutical service networks.

Ağrı city center was selected as the empirical application area because it represents a medium-sized urban environment characterized by heterogeneous settlement patterns, centralized healthcare infrastructure, peripheral accessibility challenges, and observable socioeconomic differences. Compared with highly complex metropolitan regions, Ağrı provides a manageable yet realistic planning environment in which accessibility inequalities, healthcare service concentration patterns, and pharmacy allocation decisions can be analyzed more transparently. Therefore, the city offers an

appropriate testing ground for evaluating the effectiveness of equity-oriented pharmacy planning frameworks.

Rather than treating urban space as an unconstrained continuous plane, this research models the urban landscape as a constrained spatial system in which physical barriers, legislative thresholds, healthcare infrastructure distribution, and socioeconomic vulnerabilities interact to determine the actual accessibility conditions of pharmaceutical services. In this context, pharmacy location planning is not approached as a simple commercial facility placement problem. Instead, it is formulated as a multidimensional healthcare accessibility problem requiring the simultaneous consideration of efficiency, equity, feasibility, and service continuity.

Unlike conventional pharmacy planning approaches that rely primarily on static population thresholds, the proposed framework evaluates pharmacy accessibility as a multidimensional socio-spatial problem. Accordingly, the model simultaneously considers accessibility efficiency, spatial equity, healthcare vulnerability, operational workload balancing, healthcare infrastructure integration, and temporal service continuity within a unified optimization structure.

Through this perspective, the study aims not only to improve average healthcare accessibility conditions but also to strengthen accessibility performance for disadvantaged and underserved peripheral populations. This is particularly important because average accessibility indicators may conceal severe inequalities experienced by remote or vulnerable communities. Therefore, the proposed framework explicitly considers both system-wide efficiency and worst-case accessibility conditions.

Although pharmacy openings and relocations in Türkiye are regulated through population-based criteria under the Regulation on Pharmacists and Pharmacies enacted in 2012, practical implementation patterns indicate significant spatial inequalities in pharmaceutical service distribution. Particularly within regional centers such as Ağrı, pharmacies naturally tend to cluster around commercially attractive central districts characterized by high pedestrian mobility, developed retail ecosystems, stronger purchasing power, and proximity to major healthcare facilities.

This market-driven centralization behavior attracts private investment toward already developed urban zones while leaving lower-density peripheral neighborhoods and socioeconomically disadvantaged sectors with relatively lower levels of

pharmaceutical accessibility despite their higher healthcare vulnerability. Consequently, the existing system may satisfy regulatory requirements while still failing to ensure equitable accessibility outcomes at the local level.

This discrepancy highlights the need for optimization-based planning frameworks capable of balancing demographic demand, spatial accessibility, healthcare equity, regulatory feasibility, and operational sustainability simultaneously. In this study, the proposed model is designed to support this objective by transforming the pharmacy location problem into a structured mathematical decision problem.

Socio-Spatial Disparities and Decision Friction

The spatial imbalance in pharmacy distribution generates several public health consequences, including reduced healthcare accessibility, longer travel distances, increased mobility burdens for vulnerable populations, uneven service utilization patterns, delayed pharmaceutical access, and broader social welfare losses. These consequences are particularly important for elderly individuals, patients with chronic diseases, disabled populations, and individuals with limited transportation opportunities.

Consequently, the pharmacy location problem cannot be interpreted merely as a commercial facility placement problem. Instead, it represents a multidimensional healthcare planning and spatial allocation problem involving competing objectives related to accessibility efficiency, spatial equity, operational sustainability, healthcare infrastructure integration, and fair service distribution.

Within this context, the central decision problem addressed in the present study concerns determining which candidate location nodes within the 103 segmented zones of Ağrı city center should be selected for pharmacy allocation in order to generate the most balanced, accessible, and operationally viable network.

The decision-making process evaluates candidate nodes according to multiple criteria, including:

- Weighted demand structures (w_i)
- Localized population density and settlement patterns
- Regional transportation accessibility conditions
- Centrality-based opening cost proxies (C_j)

- Distance relationships between demand points and candidate pharmacy locations (d_{ij})
- Proximity to primary healthcare facilities
- Accessibility protection requirements
- Operational balancing conditions

The strategic challenge lies in selecting an optimal facility configuration from a finite discrete candidate set while simultaneously satisfying municipal planning constraints, healthcare accessibility objectives, regulatory requirements, and economic feasibility considerations. Since each candidate location has different spatial, accessibility, cost, and healthcare proximity characteristics, the problem requires a systematic optimization approach rather than intuitive or rule-based decision making.

The optimization framework seeks to ensure that pharmaceutical services remain accessible to all population groups while minimizing excessive spatial concentration and preventing service gaps across underserved regions. Accordingly, the study adopts a hybrid planning perspective that balances efficiency-oriented, equity-oriented, cost-oriented, and healthcare-oriented objectives within a mathematically integrated framework.

Operational Balancing and Dynamic Layering

From an operational perspective, pharmacy location decisions significantly influence healthcare continuity, service quality, and the efficiency of pharmaceutical distribution systems. Properly positioned pharmacies reduce travel times for individuals requiring frequent pharmaceutical access, particularly elderly individuals, chronically ill patients, disabled populations, and individuals with limited transportation mobility.

More balanced spatial distributions therefore contribute not only to improved individual accessibility but also to broader public health performance, healthcare system resilience, and service continuity. A pharmacy network that is spatially balanced can reduce excessive dependence on central locations, improve access for peripheral neighborhoods, and strengthen the overall responsiveness of the healthcare system.

The decision problem addressed in this study also incorporates service balancing considerations. Uneven pharmacy distributions may create excessive demand

accumulation within certain facilities while leaving others underutilized. Such imbalances may negatively affect healthcare accessibility quality, waiting conditions, operational performance, and service continuity. Therefore, the proposed model does not only determine where pharmacies should be located, but also considers how demand should be distributed among selected facilities.

Accordingly, the proposed framework enforces lower capacity limits ($L=1.0$) and upper capacity bounds ($U=4.0$) to govern the balanced allocation of demand among selected pharmacies. These capacity bounds are used to prevent unrealistic service allocation patterns and to support a more balanced distribution of workload across the pharmacy network.

Another important dimension of the problem concerns logistical and operational sustainability. Pharmacies operate within pharmaceutical supply chains that require regular product distribution, inventory continuity, and efficient transportation structures. Consequently, the spatial relationship between pharmacies, healthcare facilities, and pharmaceutical distribution networks significantly affects overall operational performance.

In addition to strategic pharmacy location-allocation decisions, the present study incorporates a dynamic duty-pharmacy scheduling layer. Beyond determining where pharmacies should be strategically located, the framework evaluates how selected pharmacies can provide balanced and accessible service continuity during nighttime, weekend, and holiday periods. This additional layer strengthens the practical relevance of the model because pharmacy accessibility is not limited to regular daytime operations.

The decision problem is formulated as a multi-objective optimization problem because healthcare accessibility planning inherently involves conflicting goals. Minimizing average accessibility distance may improve aggregate system efficiency but may simultaneously neglect geographically disadvantaged peripheral areas. Conversely, prioritizing accessibility equity alone may increase operational and logistical costs.

Therefore, the study adopts a hybrid optimization structure capable of balancing these competing planning priorities in a transparent and systematic manner. To address this multidimensional decision environment, the proposed framework integrates

classical facility location theory, weighted accessibility structures, vulnerability-sensitive planning principles, Geographic Information Systems (GIS)-supported spatial analytics, and Mixed Integer Linear Programming (MILP)-based optimization techniques within a unified analytical structure.

3.2. Detailed System Boundary and Objectives

The pharmacy location-allocation problem addressed in this research consists of a multidimensional healthcare accessibility structure involving spatial, demographic, socioeconomic, operational, and logistical components. The problem is formulated within the geographical planning boundary of Ağrı city center and seeks to determine the most appropriate pharmacy distribution structure capable of balancing accessibility efficiency, spatial equity, operational sustainability, healthcare infrastructure integration, and service continuity simultaneously.

The analytical environment considered within the scope of this study represents the complete urban footprint of Ağrı city center. Unlike traditional pharmacy planning approaches that rely solely on population thresholds, the proposed framework models the city as an interconnected accessibility network consisting of demand points, candidate pharmacy locations, healthcare infrastructure nodes, transportation relationships, and operational service layers.

The system is designed to support both strategic and operational decision-making processes. Strategic decisions determine where pharmacies should be located and how demand should be assigned, whereas operational decisions determine how selected pharmacies should provide emergency and duty services throughout the planning horizon.

The framework therefore extends beyond a conventional facility placement problem and instead represents an integrated healthcare accessibility planning system in which multiple decision layers interact simultaneously.

3.2.1. System Components

The system considered within the scope of the study comprises 103 interconnected demand and candidate location zones ($I = J = 103$) representing the full urban footprint of Ağrı city center. Each zone functions simultaneously as a demand node and a potential candidate location for pharmacy establishment.

The optimization framework incorporates the following fundamental system components:

Geographically Distributed Demand Nodes (I)

The demand layer consists of 103 segmented urban zones representing localized population clusters distributed throughout Ağrı city center. Each demand node captures the healthcare service requirements of its surrounding population and serves as the primary source of pharmaceutical demand within the model.

These demand nodes are spatially distributed across the study area and reflect the heterogeneous demographic and settlement characteristics of the city. Consequently, healthcare demand is not assumed to be uniformly distributed but varies according to local population conditions.

Vulnerability-Sensitive Demand Matrix (w_i)

To represent differences in healthcare needs across neighborhoods, the study employs a vulnerability-sensitive demand structure. Demand values are generated using population-based indicators and adjusted through a demand intensity factor (demand scale = 1.5).

The weighting structure incorporates:

- Population density
- Elderly population ratio
- Healthcare dependency indicators
- Localized demand intensity

Through this mechanism, neighborhoods exhibiting greater healthcare vulnerability exert stronger influence within the optimization process. Consequently, the framework moves beyond purely population-based planning and incorporates healthcare need intensity directly into the decision-making structure.

Candidate Location Matrix (J)

The candidate location set consists of 103 potential pharmacy locations distributed throughout Ağrı city center. Each candidate node represents a feasible location at which a pharmacy may potentially be established. The optimization model evaluates

alternative combinations of these candidate locations and determines which subset should be selected to generate the most balanced accessibility network.

Geographic Spatial Filtering Layer (Allowed Location)

One of the distinctive features of the proposed framework is the incorporation of a geographic preprocessing layer. Unlike many classical location-allocation studies, the model explicitly considers land-use restrictions and spatial feasibility conditions.

Restricted urban zones are removed before optimization begins. In particular, the military areas associated with the 12th Mechanized Infantry Brigade are excluded from the candidate set because pharmacy establishment is not legally or operationally feasible within these regions.

This preprocessing stage ensures that all candidate locations evaluated by the optimization model remain practically implementable.

Primary Healthcare Infrastructure Nodes (H)

The healthcare infrastructure layer consists of major healthcare facilities operating within Ağrı city center.

These facilities include:

- Ağrı Training and Research Hospital
- Ağrı Oral and Dental Health Center
- Central Community Health Center
- Other major healthcare service providers included within the dataset

Because pharmacies frequently operate in close interaction with healthcare institutions, the proposed framework explicitly incorporates healthcare facility proximity into the optimization process. This enables the model to promote clinical synergy between pharmacies and primary healthcare infrastructure.

Transportation-Based Distance Matrix (d_{ij})

Accessibility relationships are represented through a transportation-based distance matrix generated using geographic coordinates.

A complete (103×103) distance matrix is constructed using Haversine-based calculations. This matrix represents the spatial separation between each demand node and each candidate pharmacy location.

The distance matrix serves as one of the primary inputs of the optimization model and directly influences accessibility calculations, assignment decisions, and equity evaluations.

Operational Service Continuity Layer (V_{jt})

Beyond strategic facility location planning, the framework incorporates an operational service layer used for duty-pharmacy scheduling.

The operational model considers a 30-day planning horizon ($T = 30$) and evaluates how selected pharmacies should provide emergency pharmaceutical services throughout the scheduling period.

This additional layer enables the framework to address both long-term accessibility planning and short-term operational continuity simultaneously.

Decision Variables

Within this framework, the central decision problem concerns determining:

- Which candidate pharmacy locations should be selected (x_j)
- How demand regions should be assigned to selected pharmacies (y_{ij})
- Which pharmacies should operate during each duty period (V_{jt})

Together, these variables define both the strategic and operational structure of the pharmaceutical service network.

3.2.2. Research Objectives

The proposed optimization framework is developed to achieve multiple healthcare planning objectives simultaneously. These objectives reflect the multidimensional nature of healthcare accessibility and represent the core performance criteria considered throughout the study.

Objective 1: Minimizing Average Accessibility Distance

The first objective seeks to minimize the weighted aggregate travel burden required to access pharmaceutical services. Reducing travel distances improves healthcare

accessibility, decreases transportation burdens, and enhances the overall efficiency of the pharmaceutical service network. This objective corresponds primarily to the efficiency-oriented component of the framework.

Objective 2: Improving Spatial Equity

The second objective seeks to minimize the worst accessibility conditions experienced within the network. Rather than focusing exclusively on average performance, the framework also protects geographically disadvantaged populations by reducing maximum accessibility burdens. This objective represents the equity-oriented dimension of the model.

Objective 3: Enforcing Economic Feasibility

The third objective incorporates proxy pharmacy opening costs into the optimization process. Centrality-based opening cost structures are used to represent economic considerations associated with pharmacy establishment decisions. Through this objective, the framework seeks to balance accessibility improvements with economic feasibility.

Objective 4: Maximizing Clinical Synergy

The fourth objective promotes proximity between pharmacies and major healthcare facilities. Because pharmacies frequently interact with hospitals and healthcare centers, encouraging spatial proximity can improve healthcare coordination, patient convenience, and overall service integration.

Objective 5: Ensuring Workload Balance

The fifth objective seeks to maintain balanced service loads throughout the pharmacy network. Lower and upper capacity bounds are imposed to prevent excessive demand accumulation within individual pharmacies and to improve operational efficiency across the system.

Objective 6: Securing Emergency Operational Continuity

The sixth objective concerns maintaining accessibility during emergency and duty-service periods. The operational scheduling layer seeks to ensure adequate duty-pharmacy coverage through distance constraints, coverage requirements, workload balancing mechanisms, and mandatory rest intervals.

Together, these objectives establish a comprehensive planning framework capable of addressing both strategic accessibility concerns and operational service continuity requirements.

3.2.3. Scope of the Optimization Framework

The pharmacy location-allocation problem addressed in this study extends beyond a conventional facility placement problem. Instead, it represents a multidimensional healthcare optimization problem involving accessibility, equity, operational feasibility, healthcare vulnerability, infrastructure integration, and service sustainability considerations.

The optimization framework incorporates:

- Real coordinate-based spatial data
- Geographic feasibility restrictions
- Vulnerability-sensitive demand structures
- Transportation-based accessibility measures
- Healthcare facility proximity indicators
- Pharmacy assignment relationships
- Capacity balancing constraints
- Sparse optimization structures
- GIS-supported visualization outputs
- Operational duty-pharmacy scheduling mechanisms

A total of 103 candidate locations and 103 demand zones are evaluated within the model. Although the complete problem contains 10,609 theoretical assignment relationships, the application of maximum accessibility constraints reduces the active assignment structure to 7,825 feasible demand-pharmacy pairs.

This sparse optimization structure significantly reduces computational complexity while preserving solution quality and practical realism.

The strategic optimization layer ultimately determines the most appropriate pharmacy allocation structure under the regulatory upper bound of ($P = 41$) pharmacies. Given the existing pharmacy infrastructure within Ağrı city center, the resulting solution corresponds to the establishment of 30 pharmacies within the optimized network configuration.

Accessibility maps, assignment structures, workload distributions, healthcare integration outputs, and service coverage analyses are subsequently generated to evaluate the performance of the proposed framework.

Ultimately, the proposed framework seeks to address healthcare accessibility challenges through the integration of optimization modeling, vulnerability-sensitive planning, GIS-supported analysis, healthcare infrastructure considerations, and operational healthcare management within a unified decision-support structure.

The following chapter presents the mathematical formulation of the proposed optimization model, including objective functions, decision variables, parameter definitions, normalization structures, accessibility constraints, assignment relationships, and operational scheduling formulations that collectively define the analytical foundation of the study.

4. METHODOLOGY

4.1. Methodological Philosophy and Research Design

This study addresses the problem of determining the most suitable locations for newly established pharmacies within the framework of healthcare facility location-allocation models. Pharmacies constitute one of the most accessible and frequently utilized components of public healthcare infrastructure; therefore, their spatial planning cannot be evaluated solely through conventional efficiency-oriented metrics. Beyond minimizing average accessibility costs, modern healthcare planning should also address structural inequalities affecting geographically disadvantaged populations.

Accordingly, the pharmacy location-allocation problem addressed in this study is formulated as a multi-objective optimization problem that integrates efficiency (P-median), equity (P-center), economic sustainability, and healthcare integration within a unified mathematical structure.

The methodological framework combines strategic facility placement, weighted demand analysis, GIS-supported spatial filtering, and operational healthcare scheduling within a two-stage analytical structure. Through this approach, the study seeks to bridge the gap between theoretical facility location models and practical healthcare accessibility planning.

4.2. Site Selection–Allocation Approach

Location-allocation models function as decision-support tools that simultaneously determine facility placement and demand assignment. Unlike traditional approaches that separate location selection from demand allocation, the proposed methodology treats these decisions as interconnected components of a unified optimization framework.

The model is structured around a discrete network system consisting of:

Demand Nodes ($I=103$)

Geographically distributed population clusters representing heterogeneous demographic and socioeconomic demand characteristics within Ağrı city center.

Candidate Pharmacy Locations ($J=103$)

A finite set of potential pharmacy locations evaluated under the regulatory capacity limit of ($P=41$).

Healthcare Infrastructure Nodes (H)

Major healthcare facilities, including Ağrı Training and Research Hospital, which serve as important healthcare attraction centers within the urban system.

Accessibility Matrix (d_{ij})

Distance matrices generated through coordinate-based Haversine calculations to represent realistic spatial accessibility relationships between demand nodes and candidate locations.

Vulnerability-Sensitive Demand Structure (w_i)

A weighted demand matrix incorporating population density and elderly population ratios through a demand scaling factor of 1.5.

Dynamic Duty Scheduling Horizon ($T=30$)

A 30-day operational planning horizon used for emergency duty-pharmacy scheduling.

4.3. P-Median Model: Efficiency-Oriented Optimization

The P-median component is employed to minimize the total weighted travel distance between demand nodes and assigned pharmacies. This component represents

the efficiency-oriented dimension of the optimization framework and is governed by the weighting coefficient ($\alpha = 0.35$).

By minimizing aggregate travel distance, the model seeks to improve overall accessibility efficiency and reduce transportation burdens across the urban system. The P-median structure naturally favors high-demand population clusters and contributes to the economic sustainability of pharmacy operations through improved accessibility performance.

4.4. P-Center Model: Equity-Oriented Accessibility Protection

Complementing the P-median structure, the P-center component seeks to minimize the maximum accessibility distance experienced by any demand node. This equity-oriented objective is represented by the weighting coefficient ($\beta = 0.34$).

The P-center formulation directly addresses the needs of geographically isolated and underserved populations by reducing worst-case accessibility conditions. Consequently, this component helps prevent the emergence of pharmacy deserts and strengthens accessibility protection for peripheral communities.

4.5. Normalized Weighted Sum Method: Multi-Criteria Integration

Healthcare accessibility planning inherently involves multiple and often conflicting objectives. Accessibility efficiency, spatial equity, economic feasibility, and healthcare infrastructure integration are measured using different units and scales. Consequently, direct aggregation of these objectives within a single mathematical formulation may produce biased optimization results because objective components with larger numerical magnitudes can dominate the solution process regardless of their actual policy importance.

To overcome this limitation, the proposed framework employs a Normalized Weighted Sum Method (NWSM). This approach transforms heterogeneous objective components into comparable dimensionless values before aggregation. Through normalization, each objective contributes proportionally to the optimization process according to predefined planning priorities rather than numerical scale differences.

Within the proposed framework, each objective component is divided by a corresponding normalization constant (N_1, N_2, N_3, N_4). These constants represent reference values used to eliminate scale effects and ensure numerical comparability

among objectives. As a result, accessibility distances, opening costs, and healthcare integration measures can be incorporated into a unified optimization structure without introducing measurement bias.

The overall objective function consists of four integrated components.

Efficiency Component ($\alpha = 0.35$)

The first component minimizes the weighted aggregate travel distance between demand nodes and assigned pharmacies. This objective corresponds to the classical P-median structure and seeks to improve overall accessibility efficiency by reducing average travel burdens throughout the urban system.

Equity Component ($\beta = 0.34$)

The second component minimizes the maximum accessibility distance experienced by any demand node. This objective corresponds to the P-center formulation and provides direct protection for geographically disadvantaged and underserved populations.

Cost Component ($\gamma = 0.01$)

The third component incorporates centrality-based facility opening cost proxies (C_j) into the optimization process. Although economic considerations receive a relatively small weight within the objective structure, their inclusion prevents the model from generating economically unrealistic solutions and contributes to practical implementability.

Healthcare Integration Component ($\delta = 0.30$)

The fourth component minimizes distance penalties associated with major healthcare infrastructure nodes. This objective promotes spatial integration between pharmacies and healthcare institutions such as hospitals and community health centers, thereby strengthening healthcare service coordination and patient convenience.

The weighting structure reflects the planning philosophy adopted in this study. Accessibility efficiency and spatial equity constitute the primary planning priorities, while healthcare infrastructure integration serves as a complementary strategic objective. Economic considerations are incorporated to preserve practical feasibility without dominating accessibility-oriented decision making.

Consequently, the normalized weighted sum framework enables policymakers and healthcare planners to explicitly evaluate trade-offs among accessibility, equity, healthcare integration, and economic feasibility within a transparent and interpretable optimization environment.

4.6. GIS-Based Spatial Accessibility Analysis

Geographic Information Systems (GIS) constitute a central component of the methodological framework and are employed throughout both the preprocessing and post-optimization stages of the study.

During the preprocessing phase, GIS-based spatial analysis is used to prepare, validate, and organize geographic datasets. Demand nodes, candidate pharmacy locations, healthcare infrastructure facilities, and accessibility relationships are represented using coordinate-based spatial information. This process establishes the geospatial foundation upon which the optimization framework operates.

A particularly important function of GIS within the proposed framework involves geographic filtering procedures. Unlike many conventional facility location studies that assume unrestricted urban space, the present study explicitly incorporates real-world land-use constraints into the planning process. Restricted urban zones are identified and excluded prior to optimization. In particular, areas occupied by the 12th Mechanized Infantry Brigade military facilities are removed from the candidate location set, ensuring that all evaluated pharmacy locations remain legally permissible and operationally feasible.

Following optimization, GIS is employed as a post-analysis and visualization tool. The resulting pharmacy allocation structure, accessibility distributions, service coverage patterns, and assignment relationships are visualized through spatial mapping procedures. These outputs allow decision-makers to interpret optimization results more effectively and evaluate accessibility performance across different parts of the city.

Accessibility conditions are visualized through Folium-based interactive maps that classify service performance into multiple accessibility categories. This approach facilitates the identification of underserved areas, accessibility bottlenecks, and spatial inequalities that may not be immediately apparent from numerical optimization outputs alone.

Consequently, GIS serves not only as a visualization platform but also as an analytical bridge connecting spatial data, optimization procedures, and policy-oriented interpretation.

4.7. Duty Pharmacy Scheduling and Operational Continuity

The strategic pharmacy location-allocation layer is complemented by a secondary operational scheduling model designed to ensure emergency pharmaceutical service continuity throughout the planning horizon.

While the first-stage optimization determines where pharmacies should be located, the second-stage model focuses on how selected pharmacies should operate during emergency service periods. This distinction is important because healthcare accessibility depends not only on facility locations but also on service availability during nights, weekends, and public holidays.

The scheduling framework operates over a 30-day planning horizon ($T = 30$) and incorporates multiple operational constraints designed to balance accessibility, service continuity, and workforce sustainability.

Fleet Deployment Constraint

Exactly ($q = 2$) pharmacies are assigned to duty service during each scheduling period. This constraint ensures continuous emergency pharmaceutical coverage while maintaining manageable operational workloads.

Spatial Dispersion Constraint

A minimum inter-pharmacy separation distance of ($m = 900$) meters is imposed to prevent excessive geographic clustering among simultaneously active duty pharmacies. This requirement improves citywide accessibility coverage and reduces service overlap.

Rest-Day Constraint

A mandatory rest interval of ($g = 1$) day is enforced between consecutive duty assignments. This constraint contributes to workforce sustainability and prevents excessive operational burden on individual pharmacies.

Temporal Demand Weighting

Demand conditions vary throughout the scheduling horizon. To represent these fluctuations, weekend periods receive a weighting factor of 1,30 while official holiday periods receive a weighting factor of 1,70. These adjustments reflect increased emergency pharmaceutical demand during periods when routine healthcare services are less available.

Collectively, these operational mechanisms contribute to balanced workload distribution, emergency preparedness, healthcare accessibility, and long-term service sustainability.

4.8. Sparse MILP Architecture and Computational Design

The proposed optimization framework is formulated as a Sparse Mixed Integer Linear Programming (MILP) model.

In a conventional location-allocation formulation, all possible demand-location assignment relationships are included within the optimization process. For the present study, a complete assignment structure involving 103 demand nodes and 103 candidate locations would generate 10,609 potential assignment pairs.

However, many of these assignments are practically unrealistic because they exceed acceptable accessibility limits. Including such relationships would unnecessarily increase computational complexity while contributing little to solution quality.

To address this issue, a maximum service radius threshold of $D_{\max} = 3000$ meter is imposed during model construction.

All demand-location pairs exceeding this threshold are removed before optimization begins. Through this preprocessing procedure, 2,784 infeasible assignment relationships are eliminated from the model.

As a result, the active assignment structure is reduced from 10,609 theoretical assignment pairs to 7,825 feasible decision variables (y_{ij}).

This sparse architecture substantially reduces memory requirements, improves solver performance, accelerates convergence, and enhances computational efficiency while preserving the practical realism of the accessibility network. Importantly, the elimination process does not compromise solution quality because only accessibility relationships that violate predefined planning limits are removed.

The resulting sparse MILP framework enables the efficient solution of a large-scale healthcare accessibility problem while maintaining mathematical rigor and operational relevance.

The integration of demographic indicators, coordinate-based distance matrices, healthcare infrastructure information, GIS-supported analytics, vulnerability-sensitive demand structures, and economic cost proxies ensures that the proposed methodology produces transparent, reproducible, and policy-relevant pharmacy planning solutions for the Ağrı urban environment.

In the computational development process, the model was refined through a sequence of prototype implementations up to Prototype 15, which served as the final base scenario for the reported results. Across these prototype experiments, the assignment matrix structure, sparse MIP formulation, feasibility constraints, and hospital-proximity restrictions were systematically revised in order to reduce unnecessary decision variables, improve solver performance, and accelerate computation while preserving the original optimization logic.

5. MATHEMATICAL FORMULATIONS

This section provides the rigorous formalization of the two-stage multi-objective mathematical programming framework engineered within this study. The optimization architecture is systematically bifurcated into two synchronized components to govern both strategic and operational planning layers respectively:

1. **The Strategic Pharmacy Location and Assignment Model:** A Sparse Mixed Integer Linear Programming (MILP) matrix that handles brick-and-mortar infrastructure deployment and neighborhood demand weight allocation concurrently.
2. **The Operational Duty Pharmacy Scheduling Model:** A multi-criteria scheduling engine that regulates temporal service continuity and emergency shift dispatch over a discrete planning horizon.

5.1. The Strategic Pharmacy Location and Assignment Model:

Sets

I: Set of demand points

J: Set of candidate pharmacy locations

Parameters

d_{ij} : The distance between point i and pharmacy j .

w_i : Weight of demand point i

C_j : j pharmacy opening cost

H_j : The distance of pharmacy j location to the nearest health facility.

P : Maximum number of pharmacies that can be opened

D_{max} : Maximum acceptable service distance

L : Minimum capacity

U : Maximum capacity

$\alpha, \beta, \gamma, \delta$: Objective function weights

The basic parameter values used in the model are as follows:

$$P = 41$$

$$D_{max} = 3000$$

$$L = 1$$

$$U = 4$$

Decision Variables

$$x_j = \begin{cases} 1, & \text{If a pharmacy opens in location } j \\ 0, & \text{Otherwise} \end{cases}$$

$$y_{ij} = \begin{cases} 1, & \text{If the point of demand is assigned to pharmacy } j \\ 0, & \text{Otherwise} \end{cases}$$

Objective Function

$$\text{Min } F = \alpha \sum_{i \in I} \sum_{j \in J} w_i d_{ij} y_{ij} + \beta Z + \gamma \sum_{j \in J} C_j x_j + \delta \sum_{j \in J} H_j x_j \quad (1)$$

The objective function consists of four main components:

- Minimizing the total weighted access distance
- Reducing the worst-case access distance
- Minimizing the costs of opening pharmacies
- Penalizing locations far from healthcare points

Constraints

i. Assignment of Demand Points

$$\sum_{j \in J} y_{ij} = 1 \quad \forall i \in I \quad (2)$$

This restriction ensures that each demand point is assigned to only one pharmacy. Thus, all demand areas are covered by the system.

ii. Appointment-Opening Relationship

$$y_{ij} \leq x_i \quad \forall i \in I, \forall j \in J \quad (3)$$

A request point can only be assigned to a pharmacy that is open. Assignments to locations that are not open are prevented.

iii. Maximum Number of Pharmacies

$$\sum_{j \in J} x_j \leq P \quad \forall i \in I \quad (4)$$

This restriction limits the total number of pharmacies that can open.

iv. Maximum Distance Definition

$$d_{ij} y_{ij} \leq Z \quad \forall i \in I, \forall j \in J \quad (5)$$

The largest assignment distance in the model is represented by the variable Z. This minimizes the worst-case scenario within the system.

v. Maximum Service Distance

$$d_{ij} y_{ij} \leq D_{max} \quad \forall i \in I, \forall j \in J \quad (6)$$

Assignments outside the defined maximum service radius are prevented.

vi. Capacity Constraints

$$Lx_j \leq \sum_{i \in I} w_i y_{ij} \leq Ux_j \quad \forall j \in J \quad (7)$$

This restriction limits the minimum and maximum demand load that each pharmacy can handle. This prevents some pharmacies from becoming excessively busy while others remain completely idle.

vii. Health Point Proximity Restriction

$$H_j x_j \leq 2000 \quad \forall j \in J \quad (8)$$

Locations that are too far from healthcare infrastructure are being prevented from being chosen as pharmacies.

viii. Variable Types

$$x_j \in \{0,1\} \quad \forall j \in J \quad (9)$$

$$y_{ij} \in \{0,1\} \quad \forall i \in I, \forall j \in J \quad (10)$$

$$Z \geq 0 \quad (11)$$

5.2. The Operational Duty Pharmacy Scheduling Model

The on-duty pharmacy scheduling model determines which pharmacies will be selected to be on duty on which days among the pharmacies opened as a result of the location model during a certain planning period. This model is the second stage decision model that works after the pharmacy location selection model. The pharmacies that are decided to open in the first stage are considered permanent, and in the second stage, daily guards are appointed among these pharmacies. The main purpose of this model is not just to select a certain number of pharmacies on duty every day.

At the same time, it is aimed for the pharmacies on duty to cover the demand points in the city, to be close to hospitals and health points, not to create unnecessary clusters in the same area, and to distribute the duty load evenly among pharmacies. Therefore, the on-duty pharmacy scheduling model; It can be considered as a combined structure of the coverage problem, facility selection problem and scheduling problem.

Sets

J*: A cluster of pharmacies opened as a result of the location model.

T : Planning days set

I: Set of demand points

K : Hospital and health center set

Here, the J^* set represents the pharmacies selected as a result of the location model in the first stage. Therefore, the on-call pharmacy scheduling model works only on pharmacies that are already open, not on all candidate pharmacies. The T set represents the days in the planning horizon, the I set represents the demand points where service is expected, and the K set represents hospitals and healthcare facilities.

Parameters

- q : Number of pharmacies to be selected on duty each day
- a_t : weight of day t
- r : Demand coverage radius
- r^H : Hospital/healthcare facility coverage radius
- g : Minimum rest day
- m : Minimum distance between selected pharmacies on duty on the same day
- d_{ij} : The distance between point i and pharmacy j
- h_{kj} : The distance between health point k and pharmacy j
- $p_{jj'}$: The distance between pharmacy j and pharmacy j'
- w_i : Demand point weight
- d_j : Average/weighted demand distance for pharmacy J
- h_j : Average health point distance for pharmacy j
- R_{min} : Minimum number of on-call duties a pharmacy must undertake
- R_{max} : Maximum number of on-call duties a pharmacy can handle
- R : Targeted average weighted on-call duty load per pharmacy
- M_1 :Penalty factor for failure to cover demand points
- M_2 :Penalty coefficient for not covering health points.
- λ_1 : Shift load balance coefficient
- λ_2 :Distance to demand points penalty coefficient
- λ_3 :Distance to health facilities penalty coefficient

The basic parameter values used in the model are as follows:

$$\begin{aligned}
 q &= 2 \\
 r &= 3000 \\
 rH &= 2500 \\
 g &= 1 \\
 m &= 900
 \end{aligned}$$

Accordingly, the model selects two on-call pharmacies each day, considering a coverage radius of 3000 meters for demand points and 2500 meters for health points. To prevent the same pharmacy from being on call on consecutive days, the minimum rest day value is set to $g = 1$. To ensure that the on-call pharmacies selected on the same day are not too close to each other, the minimum distance value is set to $m = 900$ meters.

Decision Variables

$$r_{jt} = \begin{cases} 1, & \text{If the pharmacy is selected as on duty on day } t \\ 0, & \text{Otherwise} \end{cases} \quad \forall j \in J^*, \forall t \in T$$

$$u_{it} = \begin{cases} 1, & \text{If point } i \text{ cannot be covered on day } t \\ 0, & \text{Otherwise} \end{cases} \quad \forall i \in I, \forall t \in T$$

$$v_{kt} = \begin{cases} 1, & \text{If the health point cannot be covered on day } t \\ 0, & \text{Otherwise} \end{cases} \quad \forall k \in K, \forall t \in T$$

Objective function

$$\begin{aligned} & \min Z_D \\ & = M_1 \sum_{t \in T} \sum_{i \in I} w_i u_{it} + M_2 \sum_{t \in T} \sum_{k \in K} v_{kt} + \lambda_1 \sum_{j \in J^*} \left(\sum_{t \in T} a_t r_{jt} - R \right)^2 + \lambda_2 \sum_{t \in T} \sum_{j \in J^*} d_j r_{jt} \\ & + \lambda_3 \sum_{t \in T} \sum_{j \in J^*} h_j r_{jt} \end{aligned} \quad (12)$$

The objective function simultaneously optimizes the following performance criteria:

- Increasing the coverage level of demand points
- Increasing the coverage level of health points

- Evenly distributing the on-call workload among pharmacies
- Reducing the access distances to demand points
- Reducing the access distances to health points

i. Daily Number of On-Duty Pharmacies Restricted

$$\sum_{j \in J^*} r_{jt} = q \quad \forall t \in T \quad (13)$$

This constraint ensures that exactly q number of on-call pharmacies are selected for each planning day. In the prototype model, since $q = 2$, two pharmacies are designated as on-call each day.

ii. Job. Demand Point Coverage Restriction

$$\sum_{j \in J^*: dij \leq r} r_{jt} + u_{it} \geq 1 \quad \forall i \in I, \forall t \in T \quad (14)$$

This constraint aims to ensure that every demand point is covered by at least one on-duty pharmacy every day.

If there is no on-duty pharmacy within the radius r of demand point i , then $u_{it}=1$, and this is penalized in the objective function. This structure prevents the model from becoming completely unsolvable. Because if it is not possible to cover all demand points on some days, the model shows the uncovered points with a penalty variable and tries to minimize the total penalty.

iii. Targeted average weighted on-call duty load per pharmacy.

$$\tilde{R} = \frac{\sum_{t \in T} a_t r_{jt}}{|J^*|} \quad (15)$$

iv. Health Center Coverage Restriction

$$\sum_{j \in J^*: hkj \leq rH} r_{jt} + v_{kt} \geq 1 \quad \forall k \in K, \forall t \in T \quad (16)$$

This constraint aims to ensure that every hospital or health point is covered by at least one on-call pharmacy every day. If there is no on-call pharmacy within the radius

r^H of health point k , then $v_{kt}=1$, and this is penalized in the objective function. This constraint is particularly important in terms of emergency access. Having on-call pharmacies near hospitals and health centers facilitates access to medication at night and in emergencies.

v. Consecutive Shift Barrier

$$\sum_{T=t}^{t+g} r_{jr} \leq 1 \quad \forall j \in J^*, \forall t = 1, \dots, |T| - g \quad (17a)$$

This constraint prevents the same pharmacy from being selected as on-call again within a short period of time. When $g = 1$, the constraint becomes as follows:

$$r_{jt} + r_{j,t+1} \leq 1 \quad \forall j \in J^*, \forall t = 1, \dots, |T| - 1 \quad (17b)$$

This prevents the same pharmacy from being on duty for two consecutive days. This restriction is important for the fair distribution of the on-call workload and the reduction of staff fatigue.

vi. Same-Day Restriction Preventing Nearby Pharmacy Selection

$$r_{jt} + r_{j't} \leq 1 \quad \forall t \in T, \forall j, j' \in J^*: p_{jj'} < m \quad (18)$$

This restriction prevents two pharmacies selected for on-duty duty on the same day from being too close to each other. If the distance between two pharmacies, $p_{jj'}$, is less than the minimum distance limit of m , these two pharmacies cannot be selected for on-duty duty together on the same day. This restriction ensures a more balanced distribution of on-duty pharmacies within the city. Thus, unnecessary service congestion is avoided in the same area, and access to on-duty pharmacies is strengthened for different neighborhoods.

vii. Shift Balance Constraint

$$R_{min} \leq \sum_{t \in T} r_{jt} \leq R_{max} \quad \forall j \in J^* \quad (19)$$

This constraint balances the total number of on-call duties each pharmacy will receive during the planning period. It prevents a pharmacy from receiving too many on-call duties or none at all. Here, R_{min} and R_{max} can be defined as follows:

$$R_{min} = \left\lfloor \frac{|T|q}{|J^*|} \right\rfloor \quad (20)$$

$$R_{max} = \left\lceil \frac{|T|q}{|J^*|} \right\rceil \quad (21)$$

This definition ensures that the total on-call workload is shared as equally as possible among the pharmacies that are open.

viii. Variable types

$$r_{jt} \in \{0,1\} \quad \forall j \in J^*, \forall t \in T \quad (22)$$

$$u_{jt} \in \{0,1\} \quad \forall i \in I, \forall t \in T \quad (23)$$

$$v_{jt} \in \{0,1\} \quad \forall k \in K, \forall t \in T \quad (24)$$

These variable types indicate that the on-call pharmacy selection and coverage status are binary decision-making processes. The variable r_{jt} represents on-call selection, while the variables u_{it} and v_{kt} represent non-coverage status.

6. RESULTS AND DISCUSSION

This chapter presents and evaluates the empirical results obtained from the proposed integrated pharmacy location-allocation and healthcare accessibility optimization framework implemented within the urban structure of Ağrı city center. The computational outputs generated through the hybrid four-component Mixed Integer Linear Programming (MILP) model, GIS-supported spatial analysis procedures, vulnerability-sensitive demand structures, and the secondary duty-pharmacy scheduling framework are systematically examined.

The generated results are evaluated across multiple performance dimensions in order to determine whether the proposed framework can improve healthcare accessibility, reduce spatial inequalities, strengthen service continuity, and support equitable pharmaceutical service distribution throughout Ağrı city center. Particular

attention is given to accessibility performance, spatial equity, service coverage, workload balancing, healthcare infrastructure integration, and operational feasibility.

Unlike purely theoretical or descriptive healthcare accessibility studies, the present research directly incorporates real-world spatial conditions into the optimization process. The framework utilizes coordinate-based geospatial information, a 103-zone demand structure, healthcare facility locations, accessibility constraints, and land-use restrictions. Consequently, the results presented in this chapter represent not only mathematical optimization outputs but also practically implementable healthcare planning solutions.

The optimization framework successfully resolved the trade-offs among four competing objective components:

- Weighted aggregate accessibility distance minimization (P-median component)
- Maximum accessibility distance minimization (P-center component)
- Facility opening cost minimization
- Healthcare infrastructure proximity optimization

The convergence of the optimization model demonstrates that healthcare accessibility planning requires balancing efficiency, equity, operational feasibility, and healthcare integration simultaneously. Single-objective optimization approaches often generate extreme solutions that favor only one planning priority. In contrast, the proposed multi-objective framework successfully generated a balanced pharmaceutical service network capable of satisfying multiple healthcare planning objectives simultaneously.

The following sections present the strategic allocation results, accessibility outcomes, GIS-supported spatial analyses, and operational scheduling performance generated by the proposed framework.

6.1. Pharmacy Location and Assignment Results

The strategic allocation engine developed within the scope of this study generated a balanced and operationally feasible pharmacy network through the simultaneous consideration of accessibility efficiency, healthcare vulnerability, service balancing requirements, and healthcare infrastructure integration.

Operating under the regulatory capacity limit of ($P = 41$), the Mixed Integer Linear Programming model systematically evaluated all feasible pharmacy allocation configurations within Ağrı city center. The optimization process ultimately identified the facility configuration that best satisfied the multi-objective planning structure developed throughout the study.

A major technical contribution of the proposed framework lies in its computational performance. Rather than evaluating all possible assignment relationships within a dense matrix structure, the model employed a Sparse MILP architecture. A complete assignment matrix involving 103 demand nodes and 103 candidate locations would generate 10,609 theoretical assignment relationships. However, many of these assignments exceed acceptable accessibility limits and therefore contribute little to practical solution quality.

To address this issue, the preprocessing layer applied a maximum service radius constraint of ($D_{max}\}=3000$) meters and simultaneously incorporated geographic feasibility restrictions. This procedure successfully removed 3,816 infeasible assignment relationships prior to optimization.

Consequently, the optimization model operated with 7,825 active binary assignment variables (y_{ij}), substantially reducing computational complexity while preserving accessibility realism and solution quality. The resulting sparse architecture enabled rapid convergence through the HiGHS optimization solver and produced a verified optimal solution within practical computational time limits.

Table 1 summarizes the computational characteristics of the optimization framework.

Table 1. Sparse MILP Computational Performance Metrics

Performance Indicator	Value
Total Theoretical Assignment Pairs	10,609
Eliminated Infeasible Assignment Pairs	3,816
Active Assignment Variables	6,793
Geographic Restriction Filtering	Applied
Healthcare-Proximity Filtering	Applied
Solver	HiGHS
Health-Distance Infeasible Candidate Locations	17

Optimization Status	Optimal Solution
Convergence Time	Less than 30 seconds

Table 1. Sparse MILP Computational Performance Metrics

The final optimized network consists of 41 active pharmacy facilities distributed throughout the study area. Considering the existing pharmacy stock within Ağrı city center, the generated solution corresponds to the strategic establishment of 30 pharmacies. This result indicates that achieving balanced accessibility conditions under the selected accessibility constraints requires a substantial expansion of the pharmaceutical service network.

From a healthcare planning perspective, this finding is particularly important because it demonstrates that regulatory compliance alone does not necessarily guarantee equitable accessibility outcomes. Although existing population-based regulations provide an upper bound for facility establishment, the optimization results indicate that significant accessibility improvements can be achieved through strategic spatial redistribution and targeted facility expansion.

The spatial allocation structure generated by the model successfully reduced excessive concentration tendencies commonly observed under market-driven facility placement mechanisms. Instead of directing new pharmacies exclusively toward commercially attractive urban centers, the optimization framework distributed facilities toward underserved and accessibility-sensitive regions characterized by greater healthcare vulnerability.

This redistribution effect was strengthened through the interaction of vulnerability-sensitive demand structures, accessibility constraints, and capacity balancing mechanisms. The lower and upper workload bounds prevented excessive concentration of demand within individual facilities and encouraged a more balanced allocation of service responsibilities throughout the network.

Another important finding concerns geographic feasibility. GIS-based validation confirmed that all selected pharmacy locations remained outside restricted military zones associated with the 12th Mechanized Infantry Brigade. Consequently, the generated solution does not contain any mathematically feasible but operationally impossible facility locations.

The resulting assignment structure produced geographically coherent service regions characterized by compact accessibility patterns and logical spatial relationships between demand points and assigned pharmacies. This outcome provides a strong foundation for the operational scheduling framework evaluated in subsequent sections.

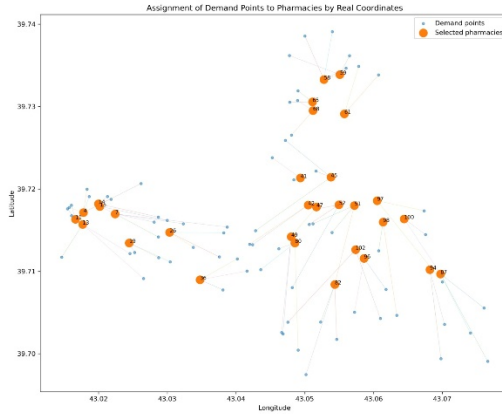


Figure 2. Spatial Assignment of Demand Points to Selected Pharmacies

6.2. Accessibility and Spatial Equity Results

One of the primary objectives of the proposed optimization framework was to improve pharmaceutical accessibility while simultaneously reducing spatial inequalities affecting geographically disadvantaged demand regions. For this reason, the performance of the generated pharmacy allocation structure was evaluated not only through average accessibility indicators but also through spatial equity measures and worst-case accessibility conditions.

Traditional healthcare planning approaches frequently emphasize aggregate accessibility improvements by minimizing average travel distances. Although such strategies may improve overall system efficiency, they often fail to protect vulnerable populations residing in peripheral neighborhoods or regions characterized by weaker transportation accessibility. Consequently, accessibility performance in the present study was evaluated from both efficiency and equity perspectives.

The empirical results demonstrate that the integration of P-median and P-center formulations generated substantial improvements in both dimensions. The combined optimization structure successfully minimized the weighted aggregate accessibility distance while simultaneously controlling maximum accessibility burdens experienced by the most disadvantaged demand regions.

A particularly important finding concerns the behavior of the maximum accessibility indicator represented by the minimax variable (Z). The optimization model successfully maintained worst-case accessibility conditions within the predefined service limits and prevented the emergence of highly isolated demand regions. This result indicates that the proposed framework effectively protected vulnerable populations from excessive travel burdens while maintaining acceptable overall efficiency.

Table 2 summarizes the principal accessibility and equity indicators generated by the optimization framework.

Indicator	Result
Demand Nodes	103
Candidate Locations	103
Total Active Pharmacies	41
Pharmacies Established	30
Total Theoretical Assignmental Pairs	10,609
Active Assignment Variables	6,793
Eliminated Infeasible Assignment Pairs	3,816
Healthcare-Proximity Feasibility Threshold	2000 m
Maximum Service Threshold (D_{max})	3000 m
Efficiency Weight (α)	0.35
Equity Weight (β)	0.34
Healthcare Integration Weight (δ)	0.30
Opening cost weight (γ)	0.01
Demand Scale Multiplier	1.5

Table 2. Accessibility and Spatial Equity Performance Indicators

One of the most important findings obtained from the accessibility analysis concerns the scale of facility expansion required to achieve equitable accessibility outcomes. The optimized network configuration activated 41 pharmacy locations, corresponding to the strategic establishment of 30 additional facilities beyond the existing pharmacy stock.

This result demonstrates that acceptable accessibility conditions could not be achieved solely through the redistribution of existing resources. Instead, substantial infrastructure expansion was required to satisfy both accessibility efficiency and equity objectives simultaneously. The findings therefore provide empirical evidence that accessibility inequalities within the study area cannot be resolved through administrative redistribution alone and may require targeted healthcare infrastructure investments.

The results further demonstrate the importance of balancing efficiency and equity objectives within a unified optimization framework. Preliminary analyses indicated that purely efficiency-oriented solutions tended to concentrate facilities within high-demand urban regions. While such configurations improved average accessibility indicators, they also generated disproportionate travel burdens for peripheral neighborhoods and geographically isolated communities.

In contrast, the hybrid optimization structure adopted in this study successfully balanced accessibility performance across the entire planning region. The near-equal weighting of efficiency ($\alpha = 0.35$) and equity ($\beta = 0.34$) enabled the framework to improve average accessibility conditions while simultaneously protecting underserved demand regions.

The vulnerability-sensitive demand structure also played a significant role in shaping the final allocation pattern. Through the application of the demand scaling factor (demand scale = 1.5), neighborhoods characterized by higher healthcare vulnerability exerted greater influence on the optimization process. As a result, demand regions associated with elderly populations, higher healthcare dependency, and weaker accessibility conditions received stronger protection within the final solution.

This finding is particularly important because conventional population-based planning approaches often assume that all demand units possess identical healthcare requirements. The proposed framework demonstrates that incorporating vulnerability-sensitive demand structures can substantially improve accessibility outcomes for populations with greater healthcare needs.

The accessibility analysis additionally revealed that the generated pharmacy network improved service continuity across peripheral regions. Several neighborhoods

previously located near the outer accessibility margins became more effectively integrated into the pharmaceutical service network. Consequently, the accessibility burden became more evenly distributed throughout the urban system.

Another significant outcome concerns healthcare infrastructure integration. The healthcare proximity component incorporated within the objective function successfully attracted pharmacy facilities toward major healthcare service nodes, particularly Ağrı Training and Research Hospital and other primary healthcare institutions. This spatial relationship improves patient convenience, reduces healthcare fragmentation, and supports continuity between diagnosis, treatment, and medication access processes.

Overall, the accessibility and spatial equity results provide strong empirical evidence that hybrid multi-objective optimization frameworks can successfully address long-standing accessibility inequalities while preserving operational feasibility and system efficiency. The findings indicate that healthcare accessibility planning benefits substantially from the simultaneous consideration of efficiency, equity, healthcare vulnerability, and infrastructure integration within a unified decision-support structure.

6.3. GIS-Based Spatial Accessibility Analysis Results

The integration of Geographic Information Systems (GIS) into the analytical framework provided important insights into the spatial performance, accessibility distribution, service coverage structure, and practical feasibility of the optimized pharmacy network. While the mathematical optimization model generated quantitative performance indicators, GIS-based analyses enabled these outputs to be interpreted spatially within the real urban environment of Ağrı city center.

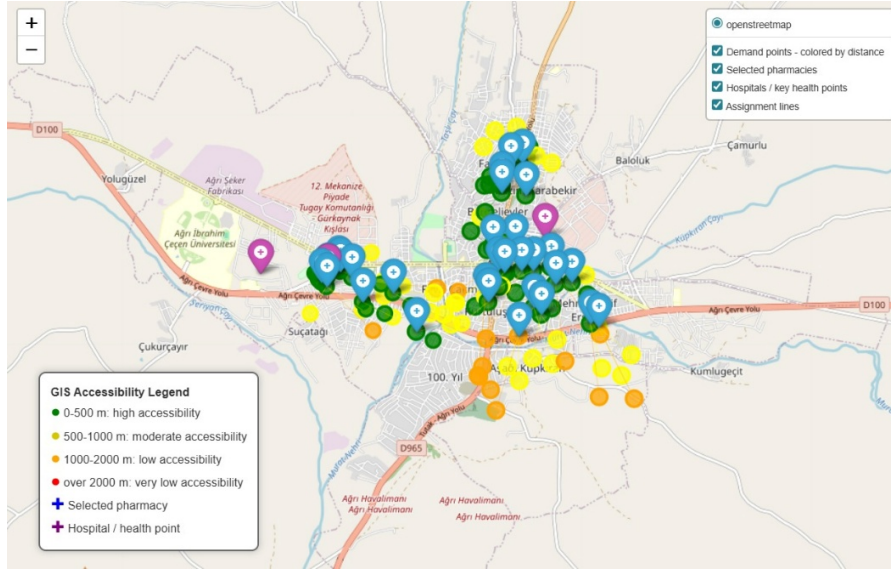


Figure 3. Optimized Pharmacy Location and Accessibility Map for Ağrı City Center

The GIS analyses demonstrated that the proposed optimization framework produced a more balanced and accessibility-oriented pharmaceutical service network compared with conventional spatial distribution patterns. By visualizing assignment relationships, service coverage zones, accessibility categories, and pharmacy locations simultaneously, GIS tools revealed the practical implications of the optimization results and supported the validation of spatial planning outcomes.

One of the most important findings concerns the reduction of excessive spatial concentration patterns. Existing pharmacy distributions within many urban environments tend to cluster around commercially attractive districts characterized by high pedestrian activity, strong transportation connectivity, and greater economic activity. Such clustering often generates redundant service coverage in some locations while leaving peripheral neighborhoods underserved.

The optimization framework substantially reduced these concentration tendencies by strategically allocating facilities toward regions exhibiting lower service density and higher healthcare vulnerability. Spatial visualizations clearly demonstrated that accessibility improvements were not restricted to central urban districts but extended throughout the broader municipal territory.

A second important finding relates to the validation of geographic feasibility. During the preprocessing stage, GIS-based spatial filtering procedures identified and excluded restricted urban areas from the candidate location set. In particular, the military zones associated with the 12th Mechanized Infantry Brigade located within

the eastern and western sections of the city were removed from the optimization environment.

Post-optimization validation confirmed that all selected pharmacy locations remained outside these restricted areas. Consequently, the resulting pharmacy network does not contain any mathematically feasible but operationally impossible facility locations. This finding strengthens the practical applicability of the proposed framework and demonstrates the importance of incorporating real-world land-use constraints into healthcare accessibility planning.

To improve interpretability, accessibility conditions were categorized into four distance-based service classes.

Near Accessibility Zone (0–600 meters)

The first category represents demand regions located within immediate walking distance of a pharmacy. Residents within this accessibility range can typically access pharmaceutical services rapidly without requiring motorized transportation. These areas represent the highest level of accessibility performance within the network.

Moderate Accessibility Zone (600–1200 meters)

The second category corresponds to standard neighborhood-level accessibility conditions. Demand regions within this range generally maintain acceptable accessibility performance and remain within comfortable walking or short-distance transportation thresholds.

Distant Accessibility Zone (1200–2400 meters)

The third category represents transitional accessibility areas where transportation support may become increasingly important. Although pharmaceutical services remain accessible, travel burdens become more noticeable compared with central accessibility zones.

Very Distant Accessibility Zone (Above 2400 meters)

The final category identifies demand regions experiencing the greatest accessibility challenges. These locations represent potential healthcare accessibility risk areas and therefore receive particular attention during the optimization process.

The GIS-based accessibility maps revealed that a substantial proportion of demand regions were concentrated within the Near Accessibility and Moderate Accessibility

categories following optimization. At the same time, the number of regions located within the Very Distant Accessibility category was significantly reduced. This redistribution indicates a more balanced accessibility structure across the study area.

Another important result concerns the spatial redistribution of accessibility burdens. Several peripheral neighborhoods that previously experienced relatively poor pharmaceutical accessibility became more effectively integrated into the healthcare service network. Consequently, accessibility performance became more evenly distributed across the city rather than being concentrated within a limited number of central districts.

Spatial Coherence Validation

An additional contribution of the GIS-supported analysis concerns the validation of spatial coherence within the generated assignment structure. The optimized service regions exhibited strong geographic continuity and compactness. Assignment relationships remained spatially logical, avoiding fragmented service territories and excessive cross-city allocation patterns that would be difficult to implement operationally.

This finding is particularly important because improvements in accessibility indicators alone do not necessarily guarantee practical applicability. A mathematically efficient solution may still generate unrealistic service territories if spatial continuity is ignored. The GIS validation process confirmed that the proposed optimization framework improved accessibility performance while simultaneously preserving realistic urban service regions.

Furthermore, the generated assignment structure demonstrated strong alignment with local transportation corridors, neighborhood boundaries, and healthcare service zones. This consistency improves operational feasibility and supports future implementation by local healthcare authorities and planning institutions.

The GIS analyses also demonstrated that the proposed framework successfully integrated pharmacies with major healthcare infrastructure elements. Several newly allocated facilities were positioned near important healthcare service providers, including Ağrı Training and Research Hospital and other primary healthcare institutions. This spatial relationship strengthens healthcare coordination, improves patient convenience, and contributes to continuity of care.

Overall, the GIS-based analyses confirm that the optimization framework not only improves numerical accessibility indicators but also generates geographically realistic, operationally feasible, and policy-relevant healthcare service structures. The spatial evidence therefore provides strong support for the practical applicability of the proposed pharmacy planning framework within the urban environment of Ağrı city center.

6.4. Pharmacy Workload Distribution and Operational Balance Results

In addition to improving healthcare accessibility and reducing spatial inequalities, the proposed optimization framework was specifically designed to support operational sustainability through balanced pharmacy workload distributions. Within pharmaceutical service systems, accessibility performance alone is insufficient if healthcare demand becomes excessively concentrated within a limited number of facilities. Uneven workload distributions may generate operational inefficiencies, longer waiting times, workforce fatigue, reduced service quality, and long-term sustainability challenges. Therefore, workload balancing constitutes a critical dimension of healthcare accessibility planning.

The empirical results demonstrate that the proposed optimization framework generated a substantially more balanced workload structure across the pharmaceutical service network. Through the integration of weighted demand assignments, vulnerability-sensitive accessibility measures, and explicit capacity constraints, healthcare demand was distributed more systematically among the selected pharmacies.

One of the most important findings concerns the reduction of excessive demand concentration within highly attractive central locations. Under conventional market-oriented pharmacy distributions, facilities located near commercial centers, transportation hubs, and major healthcare institutions frequently experience disproportionately high service intensity. Such concentration patterns create operational bottlenecks while simultaneously leaving peripheral pharmacies underutilized.

The optimization framework successfully mitigated these imbalances through the interaction of accessibility objectives and workload balancing constraints. The lower capacity threshold ($L=1.0$) and upper capacity threshold ($U=4.0$) prevented excessive

concentration of healthcare demand within individual pharmacies. These constraints ensured that service responsibilities were distributed more evenly throughout the network and prevented the emergence of highly overloaded facilities.

Furthermore, the vulnerability-sensitive demand structure strengthened workload stability. By incorporating the demand scaling factor (demand scale = 1.5), healthcare demand originating from accessibility-sensitive neighborhoods exerted greater influence within the allocation process. Rather than concentrating vulnerable populations around a small number of facilities, the model distributed demand more proportionally across the selected pharmacy network.

Histogram-based workload analyses confirmed that the generated assignment structure reduced variability among pharmacies. The final network exhibited fewer extreme workload values and a more homogeneous service distribution compared to conventional allocation patterns. This outcome contributes directly to operational sustainability because balanced utilization reduces workforce stress, improves service consistency, and enhances long-term operational resilience.

An important observation emerging from the workload analysis is that accessibility improvements were achieved without imposing excessive burdens on newly allocated pharmacies. The model successfully balanced accessibility enhancement and operational feasibility simultaneously. Consequently, the results demonstrate that accessibility equity and workload sustainability are not necessarily conflicting objectives when incorporated within an integrated optimization framework.

From a healthcare operations perspective, these findings indicate that strategic facility location decisions play a critical role not only in improving accessibility but also in ensuring efficient and sustainable service delivery. The generated workload structure therefore strengthens both healthcare accessibility and operational resilience throughout the pharmaceutical service network.

6.5. Duty Pharmacy Accessibility and Operational Continuity Results

The strategic pharmacy location-allocation framework developed in this study was complemented by a secondary operational scheduling model designed to ensure emergency pharmaceutical service continuity. While accessibility during standard operating hours is important, healthcare accessibility also depends on the availability

of pharmaceutical services during nights, weekends, public holidays, and emergency periods.

For this reason, the proposed framework incorporated a dedicated duty-pharmacy scheduling component that evaluated operational continuity across a 30-day planning horizon. This second-stage model extends the scope of traditional facility location studies by integrating strategic accessibility planning with operational healthcare service management.

The results demonstrate that the optimized pharmacy network substantially improved accessibility conditions during emergency service periods. GIS-supported accessibility evaluations indicated that the generated duty-pharmacy structure reduced operational accessibility gaps affecting peripheral and transportation-disadvantaged neighborhoods.

One of the most important findings concerns the reduction of accessibility vulnerability during non-standard service hours. Under conventional duty-pharmacy systems, emergency pharmaceutical access frequently becomes concentrated within a limited number of central locations, creating substantial travel burdens for residents of peripheral neighborhoods. The optimized scheduling framework successfully mitigated this issue by distributing emergency service responsibilities more strategically across the city.

The operational scheduling model generated the following outcomes:

- Exactly two active-duty pharmacies ($q=2$) were assigned during each operational period.
- A neighborhood coverage radius of 2500 meters ($r = 2500$) was maintained throughout the planning horizon.
- A healthcare facility protection radius of 2500 meters ($r^H = 2500$) ensured adequate emergency coverage around major healthcare institutions.
- A minimum inter-pharmacy separation distance of 900 meters ($m=900$) prevented excessive geographic clustering among simultaneously active-duty pharmacies.
- A mandatory one-day rest period ($g=1$) prevented consecutive duty assignments and supported workforce sustainability.

The results further indicate that the integration of temporal demand adjustments improved the responsiveness of the scheduling system. Weekend periods were weighted by a factor of 1,30 while official holiday periods received a weighting factor of 1,70. These adjustments enabled the scheduling framework to allocate resources more effectively during periods of elevated pharmaceutical demand.

Day	Demand Coverage	Weighted Demand Coverage	Maximum Nearest Duty-Pharmacy	Average Nearest Duty-Pharmacy Distance (m)	Uncovered Demand Points	Uncovered Hospitals	Maximum Distance (km)	Average Distance (km)	Coverage Status
1	100,00%	100,00%	3.398,90	1.188,57	0	0	3,40	1,19	Fully covered
2	100,00%	100,00%	3.073,05	1.105,96	0	0	3,07	1,11	Fully covered
3	100,00%	100,00%	2.968,26	1.136,95	0	0	2,97	1,14	Fully covered
4	100,00%	100,00%	2.979,97	1.086,90	0	0	2,98	1,09	Fully covered
5	100,00%	100,00%	2.889,49	1.153,03	0	0	2,89	1,15	Fully covered
6	100,00%	100,00%	3.159,37	1.193,60	0	0	3,16	1,19	Fully covered
7	100,00%	100,00%	2.809,18	1.096,72	0	0	2,81	1,10	Fully covered
8	100,00%	100,00%	2.678,70	1.115,27	0	0	2,68	1,12	Fully covered
9	100,00%	100,00%	2.568,21	1.217,34	0	0	2,57	1,22	Fully covered
10	100,00%	100,00%	2.956,33	1.151,39	0	0	2,96	1,15	Fully covered
11	100,00%	100,00%	3.785,82	1.418,46	0	0	3,79	1,42	Fully covered
12	100,00%	100,00%	3.410,93	1.224,45	0	0	3,41	1,22	Fully covered
13	100,00%	100,00%	2.692,66	1.220,21	0	0	2,69	1,22	Fully covered
14	100,00%	100,00%	3.089,71	1.167,94	0	0	3,09	1,17	Fully covered
15	100,00%	100,00%	2.760,08	1.315,83	0	0	2,76	1,32	Fully covered
16	100,00%	100,00%	3.515,06	1.380,31	0	0	3,52	1,38	Fully covered
17	100,00%	100,00%	3.745,88	1.327,73	0	1	3,75	1,33	Check coverage
18	100,00%	100,00%	3.757,86	1.704,59	0	2	3,76	1,70	Check coverage
19	100,00%	100,00%	3.745,88	1.386,10	0	1	3,75	1,39	Check coverage
20	100,00%	100,00%	3.398,90	1.536,55	0	2	3,40	1,54	Check coverage
21	100,00%	100,00%	2.968,26	1.390,85	0	2	2,97	1,39	Check coverage
22	100,00%	100,00%	3.145,11	1.432,93	0	2	3,15	1,43	Check coverage
23	100,00%	100,00%	3.240,22	1.439,50	0	2	3,24	1,44	Check coverage
24	100,00%	100,00%	2.908,03	1.427,64	0	2	2,91	1,43	Check coverage
25	100,00%	100,00%	3.520,01	1.560,52	0	2	3,52	1,56	Check coverage
26	100,00%	100,00%	3.478,46	1.582,36	0	2	3,48	1,58	Check coverage
27	100,00%	100,00%	3.656,93	1.522,95	0	2	3,66	1,52	Check coverage
28	100,00%	100,00%	3.760,93	1.580,51	0	2	3,76	1,58	Check coverage
29	99,03%	99,25%	4.026,34	1.896,89	1	2	4,03	1,90	Check coverage
30	93,20%	94,58%	4.291,94	2.060,37	7	2	4,29	2,06	Check coverage

Another important outcome concerns service continuity around major healthcare institutions. The duty-pharmacy network maintained strong accessibility relationships with key healthcare providers, including Ağrı Training and Research Hospital and other primary healthcare facilities. This spatial integration contributes to patient convenience and improves continuity between emergency medical treatment and pharmaceutical access.

Overall, the duty-pharmacy scheduling results demonstrate that healthcare accessibility should not be evaluated solely through static location decisions. Continuous pharmaceutical service provision requires the integration of strategic location planning and operational scheduling mechanisms. The findings therefore highlight the importance of combining facility location optimization and emergency service management within a unified healthcare planning framework.

6.6. General Discussion

The findings obtained from the proposed optimization framework demonstrate that pharmacy location planning constitutes a multidimensional healthcare planning problem that extends significantly beyond traditional population-based regulatory approaches. By integrating facility location theory, multi-objective optimization, vulnerability-sensitive demand structures, GIS-supported spatial analysis, and operational scheduling mechanisms, the study developed a comprehensive decision-support framework capable of addressing multiple dimensions of healthcare accessibility simultaneously.

One of the most important findings concerns the relationship between accessibility efficiency and spatial equity. Conventional pharmacy planning systems frequently prioritize average accessibility improvements or administrative compliance with population thresholds. While such approaches may improve aggregate performance indicators, they often fail to address accessibility inequalities experienced by geographically isolated and socioeconomically vulnerable populations.

The results obtained from the hybrid P-median–P-center framework demonstrate that efficiency and equity should not be interpreted as mutually exclusive planning objectives. Instead, the findings indicate that significant improvements in accessibility equity can be achieved while maintaining acceptable levels of operational efficiency and service sustainability. The balanced weighting structure employed in this study successfully prevented excessive prioritization of either objective and generated a more inclusive pharmaceutical service network.

This finding supports a growing body of healthcare accessibility literature arguing that average-distance indicators alone are insufficient for evaluating healthcare accessibility performance. Similar to the observations reported by Murray (2015), Pednekar and Peterson (2018), Chen et al. (2024), and Han et al. (2025), the present study demonstrates that vulnerable populations often remain underserved when planning decisions rely exclusively on aggregate efficiency indicators. Consequently, accessibility protection mechanisms must be incorporated directly into healthcare planning frameworks rather than being treated as secondary evaluation criteria.

Another important contribution concerns the integration of vulnerability-sensitive demand structures within the optimization process. Traditional accessibility models

frequently assume homogeneous demand patterns and identical healthcare needs across all population groups. However, healthcare accessibility requirements vary considerably according to demographic composition, mobility limitations, socioeconomic conditions, and healthcare dependency levels.

The application of the demand scaling mechanism within the proposed framework enabled vulnerable neighborhoods to exert greater influence on location-allocation decisions. As a result, the optimization process became more responsive to actual healthcare needs rather than relying solely on population size. This finding highlights the importance of incorporating healthcare vulnerability indicators into future healthcare accessibility models and policy frameworks.

The study also demonstrates the value of integrating healthcare infrastructure proximity into pharmacy planning decisions. Rather than evaluating pharmacies as isolated commercial entities, the proposed framework explicitly recognized their functional relationship with hospitals, family health centers, and other healthcare institutions. The resulting allocation structure generated stronger spatial integration between pharmaceutical services and healthcare providers, contributing to improved treatment continuity and healthcare coordination.

From an operational perspective, the workload balancing and duty-pharmacy scheduling results further strengthen the practical relevance of the proposed framework. The findings indicate that accessibility improvements do not necessarily require sacrificing operational sustainability. Through the use of explicit workload constraints and scheduling mechanisms, the model successfully balanced healthcare accessibility objectives with long-term service continuity requirements.

The successful implementation of the secondary duty-pharmacy scheduling layer represents another important methodological contribution. Most pharmacy planning studies focus exclusively on strategic location decisions and rarely address operational accessibility conditions during emergency periods. The results obtained in this study demonstrate that strategic facility placement and operational service continuity should be evaluated as interconnected components of a unified healthcare accessibility system.

The GIS-supported analyses provide additional evidence regarding the practical applicability of the proposed framework. Spatial visualizations confirmed that the

optimization model generated geographically coherent service regions, reduced excessive concentration patterns, and maintained compatibility with real-world land-use restrictions. The successful exclusion of military zones and restricted urban areas further demonstrates the importance of incorporating geographic feasibility considerations into healthcare accessibility planning.

From a Turkish healthcare policy perspective, the findings suggest that population-based regulatory mechanisms may be insufficient for addressing multidimensional accessibility inequalities. Although existing regulations contribute to controlling excessive pharmacy concentration, they do not explicitly account for healthcare vulnerability, transportation accessibility, demographic heterogeneity, or spatial equity considerations. Consequently, compliance with regulatory population thresholds does not necessarily guarantee equitable pharmaceutical accessibility outcomes.

The present study therefore supports the development of evidence-based and optimization-supported planning approaches capable of complementing existing regulatory frameworks. Rather than replacing current regulations, optimization-based decision-support systems can provide healthcare authorities with additional analytical tools for identifying underserved regions, evaluating alternative allocation scenarios, and improving accessibility outcomes.

An additional contribution of the study lies in demonstrating the practical applicability of sparse optimization architectures within healthcare accessibility planning. The successful reduction of theoretical assignment relationships from 10,609 to 7,825 active decision variables significantly improved computational efficiency while preserving solution quality. This finding indicates that large-scale healthcare accessibility problems can be solved efficiently through carefully designed sparse mathematical programming structures.

Overall, the findings obtained from the proposed framework provide strong evidence that pharmacy accessibility planning should be approached as a multidimensional healthcare optimization problem rather than a purely administrative or market-driven allocation process. The integration of accessibility efficiency, spatial equity, healthcare vulnerability, operational sustainability, healthcare infrastructure coordination, and emergency service continuity enables the development of more resilient and socially responsive pharmaceutical service networks.

Ultimately, the study demonstrates that hybrid optimization frameworks supported by GIS analytics and operational planning mechanisms can serve as powerful decision-support tools for healthcare authorities seeking to improve pharmaceutical accessibility within heterogeneous urban environments. The proposed framework therefore contributes both methodologically and practically to the development of more equitable, efficient, and sustainable healthcare accessibility systems.

7. CONCLUSION AND FUTURE DIRECTIONS

This study addressed the problem of pharmacy location planning and healthcare accessibility optimization within the urban environment of Ağrı city center. Unlike conventional population-based planning approaches, the research adopted a prescriptive and optimization-oriented perspective, treating pharmacy planning as a multidimensional healthcare accessibility problem involving efficiency, equity, operational sustainability, and healthcare infrastructure integration simultaneously.

To achieve this objective, a hybrid optimization framework combining classical facility location theory, Geographic Information Systems (GIS), vulnerability-sensitive demand structures, and Mixed Integer Linear Programming (MILP) techniques was developed and implemented. The proposed framework integrated four objective components within a normalized multi-objective structure: accessibility efficiency, spatial equity, opening cost considerations, and healthcare infrastructure proximity.

The empirical findings demonstrate that pharmacy accessibility planning cannot be adequately evaluated through population thresholds or average-distance indicators alone. Healthcare accessibility is influenced by multiple interacting factors, including demographic structure, transportation accessibility, healthcare vulnerability, spatial inequality, and operational service continuity. Consequently, planning approaches based solely on administrative regulations may fail to capture the true accessibility conditions experienced by different population groups.

One of the principal contributions of the study lies in the successful integration of efficiency-oriented and equity-oriented objectives within a unified optimization framework. The hybrid P-median–P-center structure demonstrated that significant improvements in accessibility equity can be achieved without sacrificing overall system efficiency. The resulting allocation structure successfully reduced accessibility

disparities affecting peripheral neighborhoods while maintaining balanced accessibility performance across the planning region.

Another important contribution concerns the incorporation of healthcare infrastructure integration into pharmacy planning decisions. By explicitly considering the spatial relationship between pharmacies and major healthcare institutions, the framework generated stronger healthcare service coordination and improved continuity between healthcare providers and pharmaceutical services.

From a computational perspective, the study demonstrated the effectiveness of sparse optimization architectures for solving large-scale healthcare accessibility problems. Through the application of a maximum service distance threshold, the optimization framework reduced the assignment structure from 10,609 theoretical demand-location relationships to 7,825 active decision variables. This reduction substantially improved computational efficiency while preserving solution quality and operational realism.

The GIS-supported spatial analysis further strengthened the practical applicability of the framework. Geographic filtering procedures successfully eliminated infeasible candidate locations located within restricted military zones, ensuring that all selected pharmacy locations remained compatible with real-world land-use conditions. As a result, the generated allocation structure was not only mathematically optimal but also geographically feasible and operationally implementable.

The final optimized network consisted of 41 active pharmacies, corresponding to the strategic allocation of 30 pharmacies beyond the existing pharmacy stock. This finding highlights the scale of accessibility improvements required to achieve equitable pharmaceutical service distribution throughout the study area and provides empirical evidence regarding the limitations of purely population-based planning mechanisms.

7.1. Limitations of the Study

Despite the methodological and practical contributions of the proposed framework, several limitations should be acknowledged.

First, the optimization model was developed under deterministic assumptions. Demand levels, accessibility conditions, travel distances, and healthcare service requirements were assumed to remain constant throughout the planning horizon.

However, real-world healthcare systems are inherently dynamic. Population growth, migration patterns, urban expansion, transportation infrastructure developments, demographic aging, and socioeconomic transformations continuously influence healthcare demand and accessibility conditions. Therefore, the generated allocation structure should be interpreted as an optimized solution under the observed conditions of the study period rather than a permanently optimal configuration.

Second, the analysis was conducted using aggregated neighborhood-level data. While this approach enabled city-scale optimization and facilitated computational tractability, it may partially conceal accessibility differences that exist within individual neighborhoods. Consequently, micro-scale variations in healthcare accessibility could not be captured fully within the current framework.

Third, although healthcare vulnerability was incorporated through weighted demand structures, additional vulnerability indicators such as disability prevalence, chronic disease intensity, income distribution, educational attainment, and healthcare utilization behavior were not explicitly included in the optimization process. The inclusion of these variables may provide a more comprehensive representation of healthcare need in future applications.

Fourth, the study focused primarily on spatial accessibility and service continuity. Behavioral dimensions of healthcare utilization, including patient preferences, pharmacy reputation, service quality perceptions, and competitive interactions among pharmacies, were not explicitly modeled. These factors may influence actual service utilization patterns even when accessibility conditions are favorable.

Fifth, the duty-pharmacy scheduling component was designed as an accessibility-oriented operational framework rather than a fully dynamic real-time scheduling system. Although the model successfully incorporates coverage, dispersion, and workforce sustainability constraints, real-time traffic conditions, emergency demand fluctuations, and short-term operational disruptions were beyond the scope of the present study.

Finally, the empirical implementation was conducted specifically within the urban environment of Ağrı city center. Although the methodological framework is transferable to other cities, local demographic characteristics, transportation structures, healthcare infrastructure distributions, and regulatory conditions may

influence optimization outcomes. Therefore, the direct numerical findings should be interpreted within the context of the selected case study.

7.2. Future Research Directions

The limitations identified in this study provide several promising directions for future research.

One important extension involves the development of dynamic and multi-period optimization models capable of representing temporal changes in healthcare demand, population distribution, transportation systems, and urban growth patterns. Such models would allow healthcare planners to evaluate long-term accessibility strategies under evolving urban conditions.

Another promising direction concerns the integration of stochastic optimization techniques. Future studies may incorporate uncertainty associated with population growth, migration movements, healthcare utilization patterns, transportation variability, and emergency demand fluctuations. This approach would enable the development of more robust healthcare accessibility frameworks capable of performing effectively under uncertain conditions.

Future research may also expand the vulnerability-sensitive planning perspective adopted in this study. Additional indicators such as chronic disease prevalence, disability rates, socioeconomic deprivation measures, healthcare utilization frequencies, and social vulnerability indices may be incorporated directly into the optimization process. Such extensions would further strengthen the equity-oriented nature of healthcare accessibility planning.

The operational component of the framework may likewise be extended. Future studies could develop real-time duty-pharmacy scheduling systems integrating traffic conditions, emergency service demand, public transportation schedules, weather conditions, and healthcare facility congestion levels. These enhancements would improve the responsiveness and resilience of pharmaceutical service networks during emergency situations.

Another important research opportunity involves incorporating behavioral and competitive dimensions into pharmacy planning models. Patient preferences, pharmacy reputation, service quality perceptions, and competitive market dynamics may be integrated through agent-based modeling, discrete choice analysis, or game-

theoretic frameworks. Such approaches would provide a more realistic representation of healthcare service utilization behavior.

Comparative applications conducted across different cities and regions of Türkiye would further strengthen the generalizability of the proposed methodology. Applying the framework to metropolitan cities, medium-sized urban centers, and rural regions would enable researchers to evaluate how demographic, geographic, and infrastructural differences influence pharmacy accessibility outcomes.

Future studies may additionally investigate the integration of pharmacy planning with broader healthcare facility location problems. Joint optimization of pharmacies, family health centers, hospitals, emergency response facilities, and pharmaceutical distribution networks may contribute to more comprehensive healthcare accessibility systems.

Overall, the proposed framework establishes a foundation for future research combining healthcare accessibility, facility location theory, spatial analytics, and operational planning. Continued development of these approaches may contribute significantly to the design of more equitable, efficient, resilient, and sustainable healthcare service systems.

7.3. Final Remarks

In conclusion, this study demonstrates that pharmacy location planning should not be viewed merely as an administrative allocation problem or a market-driven investment decision. Rather, pharmaceutical accessibility represents a multidimensional healthcare planning challenge requiring the simultaneous consideration of accessibility efficiency, spatial equity, healthcare vulnerability, operational sustainability, and service continuity.

By integrating facility location theory, multi-objective optimization, GIS-supported spatial analysis, vulnerability-sensitive demand structures, and operational scheduling mechanisms within a unified framework, the study provides an analytically rigorous and practically applicable decision-support tool for healthcare planning. The findings indicate that optimization-based approaches can play a significant role in reducing accessibility inequalities and supporting more balanced pharmaceutical service distribution across heterogeneous urban environments.

Ultimately, the proposed framework contributes both to the healthcare accessibility literature and to practical healthcare planning by demonstrating how data-driven and optimization-supported decision-making can support the development of more equitable, accessible, and sustainable pharmaceutical service networks in Türkiye.

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